S2G2HAD: A Graph-Guided Siamese Reconstruction Network for Hyperspectral Anomaly Detection

Dan Ma, Yonghong Hou, *Member, IEEE*, Minghao Chen, Beichen Li, Zhipeng Wang, and Menglong Li

***Abstract*— Hyperspectral anomaly detection (HAD) aims to identify anomalous pixels in the image with significant spectral differences from their surrounding background pixels and has important military and civilian applications. However, challenges such as high spectral similarity between pixels, data redundancy, and lack of prior information pose significant difficulties in HAD. To address these issues, we propose the selective Siamese graph-guided HAD method. In the initial phase of this study, we present a spatial–spectral feature dynamic composition module. Guided by the principles of three-way clustering theories, this module excels in achieving heightened precision in feature embedding and graph construction. Subsequently, we introduce the Characteristic Expression Differentiation mechanism, designed to enhance the separation of hidden layer features for heterogeneous data in high-dimensional space by incorporating Siamese network-derived similarity discrimination principles. Finally, we develop a graph-guided Siamese selective reconstruction module that places a strong emphasis on the differentiation between background and anomaly features. It utilizes background graph data to construct a pure background and concurrently establishes connections across spectral bands. This approach significantly enhances data processing efficiency while reducing computational resource consumption through the elimination of redundancy. Extensive experiments on seven public datasets demonstrate the effectiveness of the method.**

***Index Terms*— Characteristic expression differentiation mechanism (CEDm), graph construction, hyperspectral anomaly detection (HAD), Siamese network, three-way clustering.**

I. Introduction

REMOTE sensing is a multilevel, multiperspective, and multidomain global observation system that has become an important means of acquiring information about Earth's resources and environment. As a cutting-edge field of remote sensing technology, hyperspectral remote sensing has broken through the limitations of visible light and human vision, enabling it to more closely approximate the physical properties of surface radiation signals. It greatly enhances human capabilities in resource development, detection, and mapping at regional and global scales, expanding human horizons. Hyperspectral anomaly detection (HAD) technology can locate anomalous targets in images by analyzing spectral differences in the absence of prior information about unknown objects. Therefore, it has become an important research topic in the field of intelligent processing and analysis of remote sensing data, providing a reference for accurate identification and analysis of targets. As the demand for object features continues to increase, the target scenes covered by hyperspectral images (HSIs) are becoming more complex, and the spectral and spatial resolutions are further improved. This poses more challenges for HAD technology, making it a hot research topic and a difficult problem.

A significant amount of current research is focused on developing efficient and precise anomaly detection (AD) algorithms and techniques, including those based on statistical methods, representation models, and deep neural networks. In the initial stages of HAD research, both domestic and international researchers utilized probabilistic statistical models for AD. The Reed–Xiaoli (RX) [1] algorithm, a milestone statistical model-based AD algorithm, assumes that the background follows a multivariate Gaussian distribution, while anomalies do not. The algorithm measures the statistical characteristic difference between the target pixel and the background by calculating their Mahalanobis distance, thereby identifying the target pixel as an anomaly or not. The RX algorithm has the advantages of good detection performance, simple mathematical models, and high real-time processing efficiency. However, the complex and variable background of HSIs makes it challenging to accurately describe their distribution characteristics with a multivariate Gaussian distribution, leading to a high false alarm rate and unstable detection performance of the RX algorithm. To address these issues, researchers have optimized the RX algorithm from various perspectives [2], [3], [4], [5], [6], [7]. For example, Molero et al. [2] proposed the local RX (LRX) algorithm that improved the problem of inaccurate parameter estimation in the RX algorithm by setting the HSI to follow a Gaussian model in the local area through a dual-window sliding strategy and, thus, improved the detection performance. Zhou et al. [4] proposed the Cluster KRX algorithm that maps the original HSI to a high-dimensional

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space using a Gaussian kernel function and detects anomalies from the perspective of pixel clustering to solve the problem of high-dimensional data being inseparable in linear spaces and to provide advantages in AD on complex images. Despite achieving good detection performance, statistical model-based AD algorithms still cannot accurately represent the complex distribution characteristics of HSIs.

To address the challenge of detecting anomalies in HSIs, researchers have devoted significant attention to representation model-based AD methods, both domestically and abroad. Li and Du [8] proposed the collaborative representation detector (CRD) that utilizes a similarity matrix to regularize the collaborative representation and detect anomalous targets. Gao et al. [9] made a significant contribution to the field of HAD by presenting a novel approach that overcomes the limitations of traditional detectors. The authors achieved highly accurate detection by independently designing a chessboard-shaped topological framework that avoids making any distribution assumptions and directly mines high-dimensional data features. Based on sparse representation theory, Yuan et al. [10] proposed the local sparsity divergence-based anomaly detector (LSDAD) that constructs a dual-window model and judges whether the tested pixel is an anomaly by calculating its sparse coefficient. Since its inception, the sparse representation model has undergone numerous improvements [11], [12], [13], including the homogeneous region-based joint sparse representation (HR-BJSR) proposed by Yang et al. [14]. This method utilizes spatial frequency information of pixels to acquire dictionary vectors with better background expression capability. However, the solution of sparse representation cannot guarantee global optimality, and to address this issue, Niu and Wang [15] proposed the low-rank representation and learned dictionary (LRRaLD) method for HAD based on low-rank and sparse matrix decomposition theory to extract global information from data. The introduction of low-rank and sparse matrix decomposition theory has led to the emergence of many related algorithms [16], [17], [18], [19] in the field of HAD. Among them, Shen et al. [20] presented an innovative approach to tackle the problem of AD by introducing a matrix decomposition method with the framework and saliency priors. However, methods based on sparse representation and low-rank sparse decomposition often ignore the spatial information of HSIs during the detection process. To account for this limitation, researchers have introduced the concept of tensors into HAD. Wang et al. [21] proposed a tensor decomposition method for separating anomalous targets in images. However, as the complexity of scenes and applications increases, the background of HSIs becomes more complex, which limits the detection performance of traditional methods.

Due to the remarkable benefits of deep learning methods in data representation, they have emerged as a research hotspot for HAD. Fu et al. [22] combined a convolutional neural network (CNN)-based denoising model with a background dictionary construction method, achieving promising detection outcomes. Zhang and Cheng [23] proposed a novel HAD framework that employs fractional Fourier transform (FrFT) and tensor-based transfer CNN. Xiao et al. [24] accomplished AD by fully extracting features from a global and local perspective using a transformational neural network. Gao et al. [25] proposed a novel blind-spot self-supervised learning network that improves the semantic feature representation of the background distribution and weakens the feature expression of anomalies. Wang et al. [26] presented a novel pixel-shuffling downsampling blind-spot reconstruction network that trained a prominent background reconstruction model in a self-supervised manner. Presently, generative spectral reconstruction-based methods have become the mainstream research direction. Jiang et al. [27] proposed a discriminative reconstruction constrained generative adversarial network for HAD (HADGAN) that utilizes an unsupervised GAN model to address the background modeling issue in high-dimensional data for the first time. Jiang et al. [28] proposed a novel semisupervised spectral learning method based on GAN to explore the background information between spectral and spatial features in HSIs. Zhong et al. [29] suggested a novel background anomaly separability feature for HAD. Fan et al. [30] enhanced the robustness of their algorithm by introducing $\ell_{2,1}$-norm and graph constraints into an autoencoder structure. Wang et al. [31] introduced a novel frequency to spectrum mapping generative adversarial network (GAN) for high-resolution spectral AD. This approach mapped the original spectrum into the FrFT domain (FrFD) and reformulated it as a mapping task, thus enhancing the deep separable features of the background and anomalies. Apart from the aforementioned algorithms, several excellent generative HAD approaches [32], [33] improve detection performance by designing more reasonable constraint conditions and network structures.

Upon careful analysis, it becomes evident that the utilization of spectral imaging data for HAD remains a formidable challenge due to the intrinsic factors of spectral imaging and the rarity of anomalies. Specifically, existing research faces the following three issues.

1) Owing to inconspicuous data features, the potential for overlap between distinct categories, and susceptibility to noise, unsupervised methods frequently encounter challenges in the precise segmentation of original data pixels into well-defined categories. In these instances, the emergence of category-ambiguous data introduces ambiguity, thereby compromising the accuracy and reliability of detection models.
2) Anomalies in hyperspectral data pose a formidable challenge for detection algorithms rooted in spectral reconstruction. Specifically, these algorithms encounter limitations in establishing efficient feature extraction and representation methods, thus impeding the clear separation of heterogeneous data within the feature space. Consequently, this limitation leads to decreased purity when reconstructing background data, ultimately falling short of achieving the desired outcomes.
3) Mitigating the computational complexity inherent to high-dimensional data poses a formidable challenge. Inefficiencies in data feature extraction and utilization lead to the proliferation of large-scale models and a deceleration in inference speeds, thereby culminating in substantial resource consumption.

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In response to the aforementioned challenges, this article introduces a novel approach known as the selective Siamese graph-guided HAD (S2G2HAD) method. In contrast to existing methods, the proposed method offers three distinctive contributions, which can be succinctly summarized as follows.

1) We present a spatial–spectral feature dynamic composition module to enhance mutual information among data elements within hyperspectral graph data. This module incorporates a three-way clustering-guided strategy to effectively separate category-ambiguous data, reducing its impact on subsequent accuracy.
2) We propose the Characteristic Expression Differentiation mechanism (CEDm), designed to bolster the feature extraction and discriminative representation capabilities of the encoder. This is achieved by integrating similarity discrimination principles derived from the Siamese network framework into the hidden layers of the network. This augmentation facilitates the distinct separation of hidden layer features from heterogeneous data within a high-dimensional space, ultimately leading to an enhancement in the purity of background spectral reconstruction.
3) We introduce a graph-guided Siamese selective reconstruction module that leverages graph data containing structural information. This module employs distinct strategies to extract and represent features from the graph data for various data categories. Emphasis is placed on distinguishing between background and anomaly feature representations, with particular attention to utilizing background graph data features to achieve a pure background reconstruction. By highlighting the interconnectedness among diverse pixel samples, connections can be simultaneously established across different spectral bands. Consequently, it effectively eliminates redundant information, thereby enhancing data processing efficiency and reducing computational resource consumption.

II. Related Works

A. Siamese Neural Network

The Siamese neural network (SNN) is a coupled structure comprising two artificial neural networks that take in two samples as input and generate their embeddings in a high-dimensional space to facilitate the comparison of their similarity. Initially developed for signature verification on U.S. checks [34], SNN has demonstrated excellent performance and scalability in measuring similarity tasks and has been widely applied in various fields [35], [36], [37], [38], [39], including natural language processing and computer vision. In particular, SNN has become one of the core technologies for many applications, especially in identity recognition. For instance, face recognition [40] has been achieved by inputting two facial images into SNN and outputting their similarity. SNN has also been used in palmprint recognition [41], where a meta Siamese network (MSN) was proposed for small sample tests using FSL. In recent times, research on identity recognition based on electrocardiograms (ECGs) has gained popularity, with Zhao et al. [42] proposing a heart rate recognition algorithm based on an SNN fusion model that balances deep and artificial features, thereby improving recognition accuracy. SNN has also been applied to image classification, with Miao et al. [43] introducing a semisupervised representation-consistent Siamese network (SS-RCSN) for remote sensing image scene classification. The proposed model uses a GAN to extract discriminative features from remote sensing images through unsupervised learning and constructs a Siamese network with representation-consistent regularization loss for scene classification. Huang et al. [44] combined SNN with image change detection tasks, proposing a novel Siamese network model, multiple attention Siamese network (MASNet), which embeds selective kernel convolution (SKConv) into the encoder of SNN to improve its feature extraction ability. The network's decoder is transformed by proposing and utilizing an attention feature fusion module (AFFM) to fuse and select image features extracted from different scales and times, thereby achieving high-resolution image change detection (HRCD). Jiang et al. [45] proposed a new semisupervised Siamese network (S3N) based on transfer learning for change detection in heterogeneous remote sensing images, where low-level and deep features are separated and transferred in different ways. SNN has also found increasing use in object tracking, with Shen et al. [46] proposing a novel tracking method that introduces the attention mechanism into SNN to increase its matching discrimination and improve its matching performance through a new method of calculating attention weights. To address the high memory cost problem, Shen et al. [47] proposed a distilled Siamese tracking framework to learn a small, fast, and accurate tracker (student) from a large Siamese tracker (teacher) by capturing key knowledge through teacher–student knowledge distillation models. In summary, SNN has proven to be a valuable tool in various fields, and its versatility and capacity for measuring similarity have made it suitable for diverse tasks, including identity recognition, image classification, change detection, and object tracking.

Siamese networks have emerged as a promising tool in HSI processing. In hyperspectral detection, Wang et al. [48] proposed a meta-learning based hyper-spectral target detection using Siamese network (MLSN) that utilizes metalearning to adapt to new detection tasks quickly. The algorithm updates learned metaknowledge by employing a dual-channel Siamese network and previous target spectra. Rao et al. [49] combined Siamese networks with transformation networks to achieve efficient supervised deep network training and target detection by utilizing existing information. In hyperspectral classification, Jia et al. [50] embedded a Siamese network into a semisupervised learning scheme that integrates an autoencoder module. The proposed approach studies information in a large amount of unlabeled data and corrects it with a limited set of labeled samples, ultimately achieving hyperspectral classification. Chen et al. [51] proposed an automatic graph learning convolutional network (Auto-GCN) that unifies graph learning and hyperspectral classification using a "network-in-network" approach. In hyperspectral change detection, Wang et al. [52] proposed an end-to-end Siamese

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CNN (SiamNet) that adopts a spectral–spatial attention (SSA-SiamNet) mechanism. The proposed SSA-SiamNet method can adaptively refine spectral–spatial features by emphasizing channels and positions with high information content while suppressing those with low information content. Song et al. [53] proposed a novel cross-spatiotemporal interaction symmetric attention network (CSANet) that effectively extracts and integrates joint spatial–spectral–temporal features of HSIs while enhancing feature recognition capability of changes. Shi et al. [54] established an end-to-end trainable hyperspectral change detection model by combining a multipath convolutional long short-term memory network with a Siamese network. Rao et al. [55] introduced Siamese networks into the realm of HAD, presenting an approach for detecting anomalies in HSIs based on a transferable Siamese network architecture. This method computes anomaly scores for target pixels in relation to their neighboring pixels, leveraging learned similarity metrics, thereby achieving the task of HAD. Furthermore, it employs a contrastive loss function rooted in spectral angle distance to augment the network's capacity for generalization.

III. Methodology

This section aims to present the proposed S2G2HAD and its implementation details. Fig. 1 presents an illustrative representation of the proposed method, which encompasses three distinct components: the spatial–spectral feature dynamic composition module, the graph-guided Siamese selective reconstruction module, and AD. In the spatial–spectral feature dynamic composition module, the attainment of effective class-ambiguous data separation is realized through the employment of an integrated three-way clustering-based-guided density-based spatial clustering of application with noise (DBSCAN) [56] algorithm. The foundational step involves the application of entropy rate superpixel segmentation (ERS) [57] to procure superpixels that encapsulate spatial–spectral information intrinsic to the image. Each discrete superpixel region is subsequently assimilated as a constituent node within a graph structure, thereby fostering the establishment of critical adjacency relationships among superpixels. This profound structural metamorphosis imbues the hyperspectral data with a graph-oriented framework. The resultant pseudolabels are harmoniously amalgamated with the graph-structured data, yielding the genesis of a comprehensive graph dataset. This dataset, thereafter, serves as the input for the graph-guided Siamese selective reconstruction module. Within selective Siamese background autoencoders (S2BAEs), the dataset undergoes meticulous reconstruction, encompassing the revitalization of attribute information and the fortification of connectivity information. Remarkably, the detection of anomalies hinges upon a discerning analysis of disparities between the original HSI and its meticulously reconstructed counterpart.

A. Spatial–Spectral Feature Dynamic Composition Module

In order to establish a dependable background spectral dataset in the absence of prior information, we introduce a spatial–spectral feature dynamic composition module in this study. This module encompasses three key components: unsupervised background category discovery, hyperspectral spatial–spectral feature graph representation, and graph dataset construction. During the unsupervised background category discovery process, the effective separation of category-ambiguous data is accomplished by employing a methodology that integrates the three-way clustering-based-guided DBSCAN algorithm. During the hyperspectral spatial–spectral feature graph representation phase, we utilize ERS techniques to acquire superpixel units that encapsulate both spatial and spectral characteristics within the image. Each superpixel region is regarded as a node within the graph structure, and interconnections among these superpixels are established. This approach transforms hyperspectral data into a graph-structured format, thereby achieving a more precise conversion in terms of feature embedding. During the graph dataset construction phase, we amalgamate pseudolabels with graph-structured data to establish a dataset imbued with structural information. This method of dataset construction adeptly captures intricate relationships inherent to hyperspectral data, which are subsequently fed into the S2BAE for processing. This pivotal step significantly bolsters the model's performance.

1) Unsupervised Background Category Discovery: DBSCAN is a density-based clustering algorithm that relies on the density distribution of samples to determine their class assignments. In HSI, anomalous targets exhibit sparsity and low probability distribution. The interclass differences between anomalous samples and background samples are considerably larger than the intraclass differences within the background samples. This data characteristic endows the DBSCAN algorithm with the capability to effectively classify both the background and anomalous classes. Owing to the limited distinguishability of spectral classification features exhibited by a small fraction of pixels in HSIs, practical partitioning often encounters insufficient information. Assigning data objects to a particular cluster under these circumstances may lead to an increased probability of misclassification, causing a decline in clustering accuracy and an elevation of decision-making risks. To tackle this issue, this study incorporates the three-way clustering theory into unsupervised background category discovery, leading to more precise clustering outcomes. In contrast to the DBSCAN algorithm that classifies pixels into either background or anomalous, the three-way clustering-based-guided DBSCAN algorithm categorizes original pixels into background, anomalous, and uncertain classes.

In the course of clustering, DBSCAN examines all neighboring points within the Eps neighborhood of each center point. A cluster is formed if the number of neighbors of a center point surpasses the minimum number of points required for clustering (MinPts). For a given input $\boldsymbol{X} \in \mathbb{R}^{hsi_m \times hsi_n \times hsi_c}$, where hsi_m, hsi_n, and hsi_c correspond to height, width, and spectral band, respectively, the clustering outcome $Z = \{C_1, C_2, \ldots, C_k\} \bigcup \{Q\}$ can be acquired under the (Eps, MinPts) condition, where $\{Q\}$ represents the anomalous class and $\{C_1, C_2, \ldots, C_k\}$ represents all classes except for the anomalous class.

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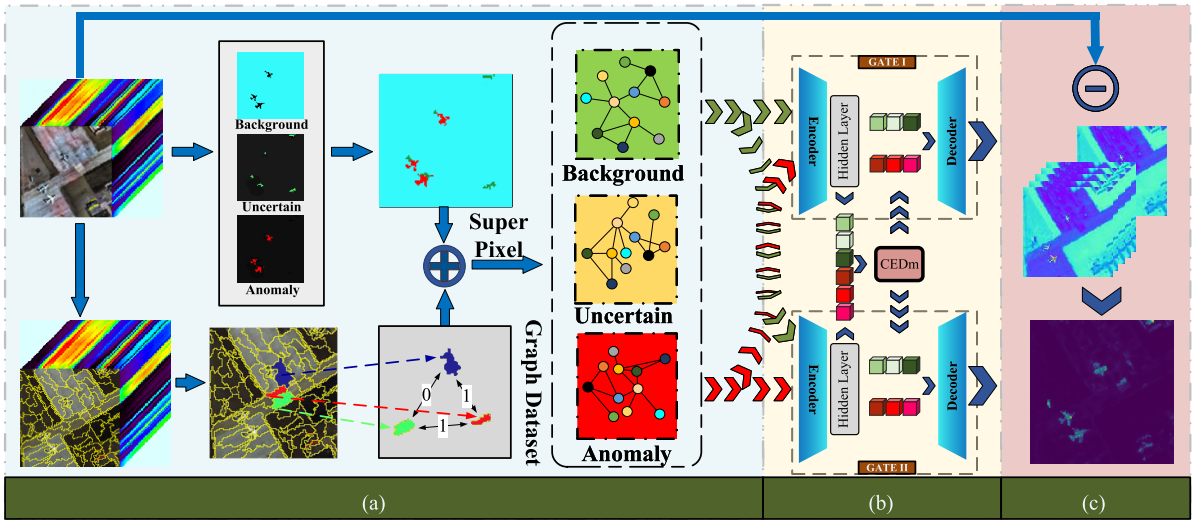


Fig. 1. Schematic of the proposed S2G2HAD. (a) Spatial–spectral feature dynamic composition module. (b) Graph-guided Siamese selective reconstruction module. (c) AD section.

The incorporation of the three-way clustering theory optimizes the hard clustering outcomes for the nonanomalous set $\{C_1, C_2, \ldots, C_k\}$. In the context of each distinct class denoted as C_j, and for any arbitrarily chosen sample $x_i \notin C_j$, it is observed that, if the natural nearest neighborhood of x_i exhibits an intersection with C_j, the logical implication arises that sample x_i merits classification within the ambit of the uncertain cluster. Similarly, when $x_i \in C_j$ and its natural nearest neighborhood is not included in C_j, x_i also belongs to the uncertain cluster. As for the distance scaling factor $\rho(\rho > 1)$, if $x_i \in C_j$, the average natural nearest-neighbor distance of x_i and the average natural nearest-neighbor distance of class C_j are computed. If $\rho\text{avg}(C_j)$ exceeds $\text{avg}(x_i)$, x_i is categorized as the background cluster; otherwise, it is classified as the uncertain cluster. The process of uncertain cluster acquisition is shown as follows:

$$\begin{cases} \text{NNN}_{nb(x_i)}(x_i) \cap C_j \neq \varnothing, & x_i \notin C_j \\ \text{NNN}_{nb(x_i)}(x_i) \not\subset C_j, & x_i \in C_j \\ \text{avg}(x_i) > \rho\text{avg}(C_j), & x_i \in C_j \end{cases} \tag{1}$$

where $\text{NNN}_{nb(x_i)}(x_i)$ refers to the natural nearest neighbor of x_i; $\text{avg}(x_i)$ represents the average natural nearest-neighbor distance of x_i, which is denoted by $\text{avg}(x_i) = (1/(nb(x_i)))\sum_{t=1}^{nb(x_i)} d(x_t, x_i)$, within this equation; and $nb(x_i)$ signifies the count of natural neighbors associated with variable x_i, whereas $d(x_t, x_i)$ conveys the spatial separation between variable x_i and its naturally nearest neighbor x_t. $\text{avg}(C_j)$ denotes the average natural nearest-neighbor distance of class C_j, denoted by $\text{avg}(C_j) = (1/|C_j|)\sum_{i=1}^{C_j} \text{avg}(x_i)$.

As a consequence, during the unsupervised background category discovery, pixel labels associated with the anomaly set $\{Q\}$ are designated as 1, thereby generating anomaly sample dataset $\boldsymbol{A}_S$. Conversely, uncertain pixels within the nonanomaly set $\{C_1, C_2, \ldots, C_k\}$ receive a label of 2, whereas background pixels are assigned a label of 0, resulting in the acquisition of uncertain sample dataset $\boldsymbol{U}_S$ and background sample dataset $\boldsymbol{B}_S$, as outlined in the following:

$$\begin{cases} \boldsymbol{A}_S = \{x_i | i \leftarrow c_{(i_1,i_2)}{}^{i_1=hsi_m, i_2=hsi_n}_{i_1=1, i_2=1} = 1 \\ \boldsymbol{B}_S = \{x_i | i \leftarrow c_{(i_1,i_2)}{}^{i_1=hsi_m, i_2=hsi_n}_{i_1=1, i_2=1} = 0 \\ \boldsymbol{U}_S = \{x_i | i \leftarrow c_{(i_1,i_2)}{}^{i_1=hsi_m, i_2=hsi_n}_{i_1=1, i_2=1} = 2 \end{cases} \tag{2}$$

where $c_{(i_1,i_2)}$ denotes the spectral vector associated with the pixel located at spatial coordinates (i_1, i_2).

2) Hyperspectral Spatial–Spectral Feature Graph Representation: The abundance of spectral information in HSIs can lead to an increase in computational complexity during node embedding and classification. Furthermore, many existing algorithms for HAD overlook the local structural information of hyperspectral data and fail to capture the correlation between hyperspectral data, which can result in a decrease in detection performance. In contrast, superpixel segmentation is a useful approach for producing homogeneous regions and extracting spatial information. Homogeneous regions tend to contain pixels with similar spectral characteristics, and each superpixel region has a smaller spatial extent and a certain spatial continuity, which can enhance the accuracy of AD.

In this study, we adopt ERS to acquire densely homogeneous and uniformly sized regions, which are utilized to learn the intrinsic low-dimensional features of various regions in HSI. This approach facilitates the generation of superpixels that represent the spatial–spectral information of the image. Each superpixel region is considered a node in the graph structure, and the adjacency relationship between superpixels is established. This representation enables the conversion of hyperspectral data into graph-structured data, thereby enabling more precise feature embedding.

Specifically, the ERS is applied to HSI $\boldsymbol{X}$. This methodology formulates the superpixel segmentation problem as an optimization problem in graph theory, wherein the image is transformed into a graph $\mathcal{G} = (V, E)$, with V representing the set of vertices and E representing the set of edges $e_{i,j}$ between vertices v_i and v_j. Edge weights $\omega_{i,j}$ denote the similarity

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between vertices v_i and v_j, which are calculated using the weight function $\omega : E \to \mathbb{R}^+ \cup \{0\}$. The problem of graph segmentation is subsequently translated into a subset selection problem, in which the objective function identifies a set of edges $A \subseteq E$ that includes all connected subgraphs in graph $\mathcal{G}' = (V, A)$. By removing certain edges from the original edge set E using the objective function, image segmentation is achieved. In the ERS method, the graph's random walk entropy rate $H(A)$ can be calculated as

$$H(A) = -\sum_i \mu_i \sum_j p_{i,j}(A) \log\big(p_{i,j}(A)\big) \tag{3}$$

where $\mu_i = (\omega_i/\omega_T)$ refers to the steady-state distribution of the random walk, ω_i represents the sum of weights of all edges passing through vertex v_i, $\omega_T = \sum_{i=1}^{|V|} \omega_i$ denotes the regularization constant, and $|V|$ denotes the number of vertices present in the graph. Furthermore, the probability of random walk $p_{i,j}(A)$ can be represented as follows:

$$p_{i,j}(A) = \begin{cases} \dfrac{\omega_{i,j}}{\omega_i}, & i \neq j,\ e_{i,j} \in A \\ 0, & i \neq j,\ e_{i,j} \notin A \\ 1 - \dfrac{\sum_{j:e_{i,j}\in A} \omega_{i,j}}{\omega_i}, & i \neq j. \end{cases} \tag{4}$$

The relationship between the objective function of ERS, the random walk entropy rate term $H(A)$, and the balance term $B(A)$ can be expressed by the following equation:

$$A^* = \max_A \ \{H(A) + \lambda B(A)\} \quad \text{s.t. } A \subseteq E \tag{5}$$

where $\lambda \geq 0$ is a weight factor that can be adjusted to balance the ratio between the random walk entropy rate term $H(A)$ and the balance term $B(A)$. The objective function is maximized to achieve superpixel segmentation of the image.

As the primary information of an HSI is often captured by the first principal component [58], this article utilizes the first principal component image as the fundamental image for superpixel segmentation. The first principal component $\boldsymbol{F}$ is obtained via PCA, and ERS superpixel segmentation is subsequently conducted on $\boldsymbol{F}$ using the following equation:

$$\boldsymbol{F} = \bigcup_{i=1}^{S} \mathrm{sp}_i \quad \text{s.t. } \mathrm{sp}_i \cap \mathrm{sp}_j = \varnothing, \quad (i \neq j) \tag{6}$$

where S indicates the total number of superpixels and $\mathrm{sp}_i (1 \leq i \leq S)$ denotes the ith superpixel.

In the context of superpixel regions that represent the spatial–spectral information of an image, each superpixel region sp_i is considered a graph node, with the adjacency relationship between superpixels established to convert HSI into an undirected graph $\boldsymbol{G_{UD}} = (\mathrm{SP}, \varepsilon)$. The vertex set SP corresponds to the obtained superpixel set, and ε represents the edges connecting superpixels, which can be represented via an adjacency matrix. The adjacency matrix $\boldsymbol{Am} \in \mathbb{R}^{S\times S}$ of graph $\boldsymbol{G_{UD}}$ is defined as

$$\boldsymbol{Am}_{i,j} = \begin{cases} 1, & \text{if } \mathrm{sp}_i \text{ and } \mathrm{sp}_j \text{ are adjacency} \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

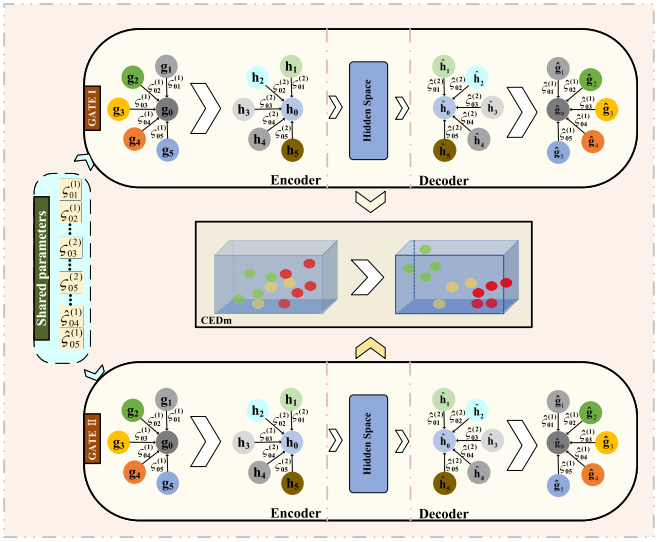


Fig. 2. Details of the graph-guided Siamese selective reconstruction module.

3) Graph Dataset Construction: During the graph dataset construction phase, the pseudolabels obtained in the unsupervised background category discovery are seamlessly integrated with the graph-structured data extracted during the hyperspectral spatial–spectral feature graph representation process. For each set of graph-structured data, outcomes acquired through the unsupervised background category discovery are meticulously linked to the superpixel graph. Throughout this procedure, we establish a feature identification threshold at 50%. In a precise manner, when a superpixel encompasses an abnormal pixel count that surpasses the threshold, it is categorized as an abnormal superpixel. Conversely, if a superpixel comprises a normal pixel count exceeding the threshold, it is classified as a normal superpixel. However, when neither category of pixel count meets the feature recognition threshold based on the superpixel's pixel count, we assign the uncertain label to that particular superpixel. Consequently, we obtain the labeled graph $\boldsymbol{G_{UD}}$, represented as $\hat{\boldsymbol{G}}_{\boldsymbol{UD}}$.

The random walk algorithm is employed to sample graph-structured data from graph $\hat{\boldsymbol{G}}_{\boldsymbol{UD}}$, resulting in the creation of dataset $\boldsymbol{G} = [\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_{N_w}] \in \mathbb{R}^{s_w \times hsi_c}$. The number of random walk sequences sampled is represented by Nw, and s_w denotes the number of steps in each random walk. In addition, the superpixel type selected for each random walk is consistent with the first superpixel type sampled.

The delineation of the construction procedure for the spatial–spectral feature dynamic composition module is elucidated through Algorithm 1, with comprehensive explication provided subsequently.

B. Graph-Guided Siamese Selective Reconstruction Module

The overall architecture of the graph-guided Siamese selective reconstruction module is depicted in detail in Fig. 2. In this module, the S2BAE is harnessed to effectuate the extraction and representation of salient features inherent within graph data. This endeavor is underpinned by a nuanced and adaptive stratagem, which tailors its approach to suit the unique characteristics of various data categories. The core objective herein

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Algorithm 1 Construction of the Spatial–Spectral Feature Composition Module

Input: (1) the original HSI $\boldsymbol{X} \in \mathbb{R}^{hsi_m \times hsi_n \times hsi_c}$;
(2) the number of superpixels S;
(3) the number of random walk sampling Nw.
Output: Constructed graph dataset
$\boldsymbol{G} = \left[\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_{N_w}\right] \in \mathbb{R}^{s_w \times hsi_c}$.
Procedure:
Obtain the clustering results $\boldsymbol{A}_S$, $\boldsymbol{B}_S$, and $\boldsymbol{U}_S$ according to Eq. (2).
Obtain superpixel segmentation region SP according to Eq. (3)(4)(5)(6).
Combining the clustering results with the $\boldsymbol{G_{UD}}$ to obtain labeled $\hat{\boldsymbol{G}}_{\boldsymbol{UD}}$.
for i $= 1, 2, \ldots, N\ w$ **do**
  Initialize an empty graph data $\boldsymbol{g}_{\mathrm{i}}$.
  Randomly select a starting superpixel and add its features to the graph data.
  for j $= 1, 2, \ldots, s_w$ **do**
    Obtain the neighborhood of the current superpixel sp_i.
    Randomly select a superpixel with the same label as the previous superpixel from the neighborhood.
    Add the features and the connection information of the selected superpixel to the graph data.
  end
end

is to proficiently distill and articulate the distinctive attributes inherent within diverse graph data typologies. In the course of delineating disparities between representations of background and anomaly features, we proactively underscore the pivotal role played by the features inherent within background graph data. Our paramount objective lies in the adept utilization of these background graph data features to meticulously fashion a pristine backdrop. By the incorporation of the CEDm within the hidden layers of the dual-graph attention autoencoder (GATE), we confer upon the encoder heightened proficiency in feature extraction and discrimination. This mechanism, in turn, enables the precise segregation of heterogeneous data within a high-dimensional space, resulting in a notable enhancement of background spectral reconstruction purity.

1) Selective Siamese Background Autoencoders: The foundational architecture of entity S2BAE comprises a pair of GATEs. The spatial–spectral feature dynamic composition module, responsible for gathering pertinent graph data, orchestrates the random allocation of this acquired information into the two constituent subnetworks within GATEs. It is imperative to emphasize that the two aforementioned subnetworks share an identical architectural configuration, underpinned by complete weight sharing, thus manifesting the absolute parameter congruity between them. Leveraging this foundational setup, we shall now embark on a comprehensive elucidation of one of these subnetworks within the overarching system S2BAE.

In the realm of graph learning, GATE leverages the self-attention mechanism to acquire knowledge of the connections and representations among nodes within a given graph. To achieve this, GATE employs a scheme for computing attention scores between each node and its respective neighbors, which serve as a proxy for the significance of each node. Subsequently, these scores are utilized to weight the nodes and generate their corresponding representations. This innovative technique endows GATE with the capability to dynamically model the relationships between distinct nodes and manifest notable generalization properties when confronted with a diverse array of graph structures.

During the process of background estimation, this study employs a method whereby the graph dataset acquired from the spatial–spectral feature dynamic composition module is inputted into the encoder of the S2BAE, thereby yielding a latent feature embedding vector. At the kth layer of the encoder, the correlation between the neighboring nodes j and i is determined via the computation outlined in the following equation:

$$e_{ij}^{(k)} = \sigma\left(\boldsymbol{v}_u^{(k)^T} \boldsymbol{W}^{(k)} \boldsymbol{h}_i^{(k-1)} + \boldsymbol{v}_r^{(k)^T} \boldsymbol{W}^{(k)} \boldsymbol{h}_j^{(k-1)}\right) \tag{8}$$

where $\sigma(\cdot)$ represents the sigmoid activation function, and $\boldsymbol{W}^{(k)} \in \mathbb{R}^{n^{(k)} \times n^{(k-1)}}$, $\boldsymbol{v}_u^{(k)} \in \mathbb{R}^{n^{(k)}}$, and $\boldsymbol{v}_r^{(k)} \in \mathbb{R}^{n^{(k)}}$ denote the trainable parameters for the kth encoder layer. $n^{(k-1)}$ and n^k signify the dimensions attributed to the $k-1$th and kth layers of the encoder. $\boldsymbol{h}_i^{(k-1)}$ and $\boldsymbol{h}_j^{(k-1)}$ correspond to the node characteristics of node i and node j within the $k-1$th encoder layer, respectively. For the initial layer, the node characteristics are equivalent to the input sample, denoted by $\boldsymbol{h}_i^{(0)} = \boldsymbol{g}_i$ $\forall i \in 1, 2, \ldots, Nw$. Subsequently, the normalized attention coefficient of the kth node can be obtained as follows:

$$\varsigma_{ij}^{(k)} = \frac{\exp\left(e_{ij}^{(k)}\right)}{\sum_{l \in \mathrm{NBR}_i} \exp\left(e_{il}^{(k)}\right)} \tag{9}$$

where NBR_i represents the set of neighbors associated with node i, specifically comprising a group of nodes, including node i itself, which is connected to node i through the adjacency matrix $\boldsymbol{Am}$. The embedding $\boldsymbol{h}_i^{(k)}$ of node i in the kth encoder layer can be expressed as follows:

$$\boldsymbol{h}_i^{(k)} = \sum_{j \in \mathrm{NBR}_i} \varsigma_{ij}^{(k)} \sigma\left(\boldsymbol{W}^{(k)} \boldsymbol{h}_j^{(k-1)}\right). \tag{10}$$

Upon traversing through the Lth encoder layer, the output of the terminal layer is used to represent the nodes and is denoted as $\boldsymbol{h}_i = \boldsymbol{h}_i^{(L)}$ $\forall i \in 1, 2, \ldots, Nw$, serving as the ultimate node representation.

This study constructs a latent feature comparator between the outputs of two encoders to gauge the similarity of encoded latent feature embedding vectors. To tackle the problem of indistinct spectral classification features in a limited number of pixels, a supplementary factor denoted as CEDm is incorporated into the latent feature comparator to amplify the feature differences between typical and anomalous samples. It is crucial to note that pixels with inherently ambiguous

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classification information, or uncertain pixels, are excluded from this process to prevent their potential assimilation into an erroneous representation space.

Ultimately, the latent feature embedding vectors are directed toward the decoder in order to generate the reconstructed attribute and connectivity information. In the interest of achieving a more accurate reconstruction background, this method introduces the concept of selective partial reconstruction into the S2BAE, whereby solely data with pseudolabels of background are utilized for background reconstruction, while anomalous and uncertain data are deliberately omitted from this process.

The decoder applied in this work features an equivalent number of layers to the encoder. Each decoder layer mirrors the structure of its associated encoder layer, thereby facilitating the reconstruction of a node's representation by leveraging the representations of its neighbors, as dictated by their respective correlations. In the kth decoder layer, the attention coefficient of node j relative to node i is expressed as follows:

$$\hat{e}_{ij}^{(k)} = \sigma\left(\hat{\boldsymbol{v}}_u^{(k)^T}\hat{\boldsymbol{W}}^{(k)}\hat{\boldsymbol{h}}_i^{(k)} + \hat{\boldsymbol{v}}_r^{(k)^T}\hat{\boldsymbol{W}}^{(k)}\hat{\boldsymbol{h}}_j^{(k)}\right) \tag{11}$$

where $\sigma(\cdot)$ represents the sigmoid activation function, alongside $\hat{\boldsymbol{W}}^{(k)} \in \mathbb{R}^{n^{(k-1)}\times n^{(k)}}$, $\hat{\boldsymbol{v}}_u^{(k)} \in \mathbb{R}^{n^{(k-1)}}$, and $\hat{\boldsymbol{v}}_r^{(k)} \in \mathbb{R}^{n^{(k-1)}}$ as the trainable parameters for the kth decoder layer, where $\hat{\boldsymbol{h}}_i^{(k)}$ and $\hat{\boldsymbol{h}}_j^{(k)}$ correspond to the node characteristics of node i and node j within the kth decoder layer, respectively. The ultimate encoder layer output, designated as $\hat{\boldsymbol{h}}_i^{(L)} = \boldsymbol{h}_i^{(L)}$, is subsequently utilized as input for the decoder, with L indicating the total number of encoder layers.

The computation for the normalized attention coefficient between adjacent node j and node i in the kth decoder layer is given as follows:

$$\hat{\zeta}_{ij}^{(k)} = \frac{\exp\left(\hat{e}_{ij}^{(k)}\right)}{\sum_{l\in \mathrm{NBR}_i}\exp\left(\hat{e}_{il}^{(k)}\right)} \tag{12}$$

where NBR_i represents the neighbors of node i and the reconstructed node representation $\hat{\boldsymbol{h}}_i^{k-1}$ at the $(k-1)$th layer of the decoder can be expressed as follows:

$$\hat{\boldsymbol{h}}_i^{(k-1)} = \sum_{j\in \mathrm{NBR}_i}\hat{\zeta}_{ij}^{(k)}\sigma\left(\hat{\boldsymbol{W}}^{(k)}\hat{\boldsymbol{h}}_j^{(k)}\right). \tag{13}$$

Upon traversing through the Lth decoder layer, the output of the last layer is utilized to represent the reconstructed node attribute, denoted as $\hat{\boldsymbol{g}}_i = \hat{\boldsymbol{h}}_i^{(0)}\ \forall i \in 1, 2, \ldots, Nw$.

2) Characteristic Expression Differentiation Mechanism: In this investigation, a twin-network paradigm is harnessed, comprising dual-image reconstruction networks. Within this framework, a Siamese network is meticulously designed, incorporating the CEDm aimed at amplifying the representation of pairs of latent feature vectors, originating from the encoding process within the Siamese network. Specifically, two networks, denoted as GATE and sharing identical structures and parameters, are constructed, with Input I and Input II serving as their respective inputs. The GATE comprises an encoder and a decoder, with the intermediate variables within the encoder and decoder referred to as hidden space features. During the training process, gradient backpropagation and network parameter updates are synchronized, with the parameters of the two encoders always remaining consistent. This ensures that the reconstruction results for identical inputs are consistently identical. The graph dataset generated by the spatial–spectral feature dynamic composition module is then fed into the encoder of the Siamese network to obtain the embedding vectors, with Input I and Input II randomly fed with normal/abnormal data. The data_flag is set to 1 when both sets of data are normal or abnormal, while it is set to 0 when one set of data is normal and the other is abnormal. Concurrently, the CEDm is constructed between the encoder outputs (i.e., the hidden space features) of the two graph reconstruction networks, via the implementation of a contrastive loss function

$$L = \frac{1}{2Nw}\sum_{i=1}^{Nw} f_D d^2 + (1-f_D)\max(\mathrm{margin} - d, 0)^2 \tag{14}$$

where margin represents the threshold value, while d denotes the feature distance between the hidden space features generated by Input I and Input II after passing through the encoder $d = \mathrm{cosineSimilarity}(\boldsymbol{h}_i, \boldsymbol{h}_j) = ((\boldsymbol{h}_i \cdot \boldsymbol{h}_j)/(\|\boldsymbol{h}_i\|_2\|\boldsymbol{h}_j\|_2))$ $\forall i, j \in 1, 2, \ldots, Nw$, with f_D being equivalent to data_flag. During the training process, when Input I and Input II share the same type, f_D is set to 1, and the second term of the formula is equal to 0. For the first term, a diminishing loss value is observed with decreasing d. In contrast, when Input I and Input II differ in type, f_D is set to 0, and the first term of the formula is equal to 0. For the second term, a declining loss value is achieved as d approaches the margin.

This study demonstrates that the utilization of a loss function minimization strategy during network training can effectively decrease the intraclass distance. However, as the interclass distance d increases to a certain magnitude, the scope of loss reduction is limited, rendering it challenging to optimize the hidden space representation of data from different classes. This suggests that expanding the interclass distance can be difficult. In addition, the presence of uncertain pseudolabels may result in the inclusion of uncertain pixels in each class, which exhibits certain similarities to the features of both classes, leading to misclassification. Consequently, blindly reducing the hidden space feature distance between pixels of the same class may pose difficulties in detecting uncertain pixels. To address this issue, this study presents an optimized contrastive loss function be formulated as

$$L_{\mathrm{CEDM}} = \frac{1}{2Nw}\sum_{i=1}^{Nw}\left(f_D d^2\right)^{1/\alpha} + \beta(1-f_D)\max(\mathrm{margin} - d, 0)^2 \tag{15}$$

where α and β are collectively referred to as the characteristic expression differentiation coefficient. The coefficient $\alpha \in N_+$ is applied to the first term of the contrastive loss function, reducing the impact of intraclass distance on the loss function and mitigating the effect of minimizing the intraclass feature distance. On the other hand, the coefficient β amplifies the impact of the second term of the loss function, enabling the network to search for larger interclass distances in the feature representation. This approach simultaneously expands the hidden space feature distance between different-class data

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while minimizing the reduction of the hidden space feature distance between same-class data, thereby avoiding further assimilation of uncertain pixels in the uncertain cluster into the incorrect category. This, in turn, reduces the reconstruction bias caused by a small number of uncertain pixels in the normal class during background reconstruction, resulting in a purer reconstructed background and a lower probability of missing abnormal pixels. Furthermore, it prevents the assimilation of misclassified background pixels with abnormal spectral characteristics in the abnormal cluster, thereby decreasing the probability of false alarms. These combined efforts contribute to the subsequent improvement of abnormal detection performance.

C. Training Loss and Anomaly Detection

Graph-structured data are composed of node features and connection information. The initial step in our analysis involves minimizing the reconstruction loss of node features. This loss is defined as the ℓ_2 loss between the input of the encoder and the output of the decoder, as illustrated in the following:

$$L_{\text{att}} = \sum_{i=1}^{Nw} \left\| \boldsymbol{g}_i - \hat{\boldsymbol{g}}_i \right\|_2 \tag{16}$$

where $\|\cdot\|_2$ represents the ℓ_2-norm, and we also evaluate the similarity of the representation of adjacent nodes in the reconstruction. The graph reconstruction loss can be defined as follows:

$$L_{\text{str}} = -\sum_{i=1}^{Nw} \sum_{j \in \text{NBR}_i} \log\left(\sigma\left(\boldsymbol{h}_i^T \boldsymbol{h}_j\right)\right) \tag{17}$$

where $\sigma(\cdot)$ represents the sigmoid activation function. The overall loss function of the network can be expressed as follows:

$$L = L_{\text{att}} + L_{\text{str}} + L_{\text{CEDM}}. \tag{18}$$

During the testing phase, the reconstructed background spectral samples may contain a small number of anomalous pixels, and relying solely on the reconstruction error for AD may not yield optimal results. By analyzing the spectral and spatial characteristics of the reconstructed HSI and following the design principles of the background representation network, it is observed that the background pixels in the reconstructed image and the original HSI are nearly identical, whereas the anomalous pixels exhibit significant differences and tend to follow the background distribution. To separate the anomalous pixels and suppress the background, we propose using the difference between the original HSI and the reconstructed image. This approach ensures the consistency of the anomalous target's shape, as the reconstructed HSI conforms more closely to the background distribution. The calculation process for the difference can be described as follows:

$$\Delta \boldsymbol{X} = |\boldsymbol{X} - \hat{\boldsymbol{X}}| \tag{19}$$

where $\boldsymbol{X}$ denotes the original HSI, $\hat{\boldsymbol{X}}$ represents the reconstructed data obtained by testing the original HSI $\boldsymbol{X}$ with the background distribution model, and $\Delta \boldsymbol{X}$ represents the difference data. Once the difference data that effectively reflect the contrast between the background and anomalies are obtained, this study utilizes the Mahalanobis distance to calculate the anomaly score of the tested pixel, thereby detecting the anomalous pixels in the difference image. The computational procedure can be presented as follows:

$$M_D(\Delta \boldsymbol{X}) = \left(\Delta \boldsymbol{x}_\phi - \boldsymbol{\mu}\right)^T \boldsymbol{\Gamma}^{-1} \left(\Delta \boldsymbol{x}_\phi - \boldsymbol{\mu}\right) \tag{20}$$

where $\Delta \boldsymbol{x}_\phi \in \mathbb{R}^{hsi_c \times 1}$ refers to the ϕth vector of $\Delta \boldsymbol{X}$, and $\boldsymbol{\mu} \in \mathbb{R}^{hsi_c \times 1}$ and $\boldsymbol{\Gamma} \in \mathbb{R}^{hsi_c \times hsi_c}$ correspond to the mean value and the covariance matrix of $\Delta \boldsymbol{X}$, respectively.

Algorithm 2 summarizes the whole procedure of the proposed S2G2HAD method.

Algorithm 2 Proposed S2G2HAD Method

Input: (1) the original HSI $\boldsymbol{X} \in \mathbb{R}^{hsi_m \times hsi_n \times hsi_c}$;
(2) the number of superpixels S;
(3) the number of random walk sampling Nw;
(4) the dimensionality of the latent vector D;
(5) the value of margin.
Output: Hyperspectral anomaly detection map.
Procedure:
Obtain the graph dataset $\boldsymbol{G} = \left[\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_{N_w}\right] \in \mathbb{R}^{s_w \times hsi_c}$ by Algorithm 1.
Initialize the network with random weights.
for *each epoch* **do**
  for *each sample batch* **do**
    Encode the graph data into latent vectors.
    Calculate the contrastive loss according to Eq. (15).
    if *the label of the graph data is "background"* **then**
      Decode the latent vectors into graph data.
      Calculate the reconstruction loss of node features and graph structure according to Eq. (16)(17).
    end
    Back-propagate the total loss Eq. (18) to update the learned parameters.
  end
end
Get the anomalous results by Eq. (19)(20).

IV. Experimental Results and Analysis

A. Experimental Datasets

To verify the effectiveness of the proposed method, we evaluated its AD performance on real hyperspectral datasets, including Airport-Beach-Urban (ABU)[1] and San Diego. The ABU dataset, which consists of 13 distinct scenes encompassing a variety of environments including airports, beaches, and urban areas, is further subdivided into 13 subdatasets for analysis. The datasets comprise various anomaly types with distinct attributes. Additional information on each dataset is presented in Table I.

[1] http://xudongkang.weebly.com/

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TABLE I
DATASET DETAILS

Datasets	Dataset Detail					
	Sensor	Resolution	Spatial Size	Bands	Anomaly Material	Anomaly Proportion
airport-1	AVIRIS	7.1m	100×100	205	Lanes	1.44%
airport-2	AVIRIS	7.1m	100×100	205	Lanes	0.87%
airport-3	AVIRIS	7.1m	100×100	205	Lanes	1.70%
airport-4	AVIRIS	3.4m	100×100	191	Aircrafts	0.60%
beach-1	AVIRIS	17.2m	150×150	188	Ship	0.08%
beach-2	AVIRIS	7.5m	100×100	193	Fishery	2.02%
beach-3	AVIRIS	4.4m	100×100	188	Ship	0.13%
beach-4	ROSIS	1.3m	150×150	102	Vehicles	0.28%
urban-1	AVIRIS	17.2m	100×100	204	Buildings	0.67%
urban-2	AVIRIS	17.2m	100×100	207	Buildings	1.55%
urban-3	AVIRIS	3.5m	100×100	191	Ships	0.52%
urban-4	AVIRIS	7.1m	100×100	205	Buildings	1.70%
urban-5	AVIRIS	7.1m	100×100	205	Buildings	2.32%
San Diego	AVIRIS	3.5m	100×100	189	Aircrafts	1.34%

TABLE II
OPTIMAL PARAMETER SETTINGS FOR THE TESTED METHODS

Methods	Parameters
LRX	$(w_{in}, w_{out})_{\text{airport scenes}} = (11, 13)$ $(w_{in}, w_{out})_{\text{beach scenes}} = (11, 21)$ $(w_{in}, w_{out})_{\text{urban scenes}} = (13, 15)$ $(w_{in}, w_{out})_{\text{San Diego}} = (37, 55)$
CRD	$(w_{in}, w_{out}) = (7, 13)$, λ=10^{-6}
AED	κ=25
FrFE	p=0.8
HADGAN	s=3, d=0.5, λ=0.5
WGAN	D=3, N=128, λ=0.9, γ=30
BASGAN	λ=0.05, n=3
RGAE	λ=10^{-2}, S=150, d=100
Proposed	S=100, D=64, Nw=400,margin=1.5

B. Experimental Setup

1) Comparison Methods: In order to ascertain the efficacy and superiority of our proposed method, we conducted a comparative analysis with nine other existing methods, namely, RX [1], LRX [2], CRD [8], FrFE [7], AED [59], HADGAN [27], WGAN [28], BASGAN [29], and RGAE [30]. The baseline algorithms evaluated in this investigation demonstrate both conventional and modern techniques for HAD. Statistical methods, including RX, LRX, AED, and FrFE, pertain to the former, while CRD is an example of the classical representation-based method. As for the latter, novel techniques, such as HADGAN, WGAN, BASGAN, and RGAE, are employed.

The parameters of all these algorithms are optimized, and the reported test results reflect their optimal performance. Table II shows the optimal parameter settings for the comparative methods. The RX algorithm requires no parameter settings, while the LRX method requires two parameters: the size of the inner window (w_{in}) and the size of the outer window (w_{out}). The CRD method has three parameters: the size of the inner window (w_{in}), the size of the outer window (w_{out}), and the regularization parameter (λ). The AED method has one parameter, which is the area parameter (κ). The FrFE method has one parameter, which is the fractional order (p). HADGAN has three parameters: the filter size (s), the degree of blurring (d), and the weighting parameter (λ). WGAN has four parameters: the depth of each subnetwork (D), the number of hidden nodes (N), and the weighting parameters (λ and γ). BASGAN has two parameters: the separability constraint weight (λ) and the number of hidden layers (n). RGAE has three parameters: the weighting parameter (λ), the number of superpixels (S), and the number of hidden layers (d). The proposed method has four parameters: the number of superpixels (S), latent vector dimension (D), random walk sampling number (Nw), and the value of margin. The details of corresponding parameter selection experiments and the effect of these parameters are conducted and analyzed in Section IV-E.

2) Evaluation Criterion: We employed commonly used evaluation metrics, namely, the 3-D receiver operating characteristic (3-D ROC) [60] curve and the area under the curve (AUC) [61], to assess the effectiveness and superiority of the proposed method.

3) Implementation Details: The proposed method is developed using the Python language (Python 3.8), and it utilizes the PyTorch deep learning framework (PyTorch 1.8). The experiments are conducted on a workstation equipped with an Intel Core i7-10700 CPU, NVIDIA 2080Ti GPU, and 32 GB of RAM. To maintain consistency, the color image data for each dataset is generated using the Hyperspectral Imaging Library in the Image Processing Toolbox of MATLAB2022. The optimization of the model is carried out using the Adam optimizer, with $\beta = (0.9, 0.999)$ as the optimizer parameters. In the experimental setup, the depths of both the encoder and the decoder are set to 2, while the initial learning rate is 1×10^{-4}. The total number of optimization epochs for the model is set to 500.

C. Detection Performance

This section presents a comprehensive evaluation of the detection performance of the proposed algorithm on the ABU and San Diego datasets, including a comparison with five typical AD methods (RX, LRX, CRD, FrFE, and AED) and four latest AD methods (HADGAN, WGAN, BASGAN, and RGAE). Figs. 3–6 depict the detection results of all ten methods, utilizing 2-D images to provide a clear visualization of the predicted anomaly regions. To quantitatively evaluate the performance of the proposed method, Tables III and IV present the AUC values of each method, with the top performance indicators highlighted in bold, the suboptimal result indicated by underlining, and the third-best result presented in italics.

Furthermore, seven commonly used hyperspectral datasets are selected to provide a comprehensive visualization of the AD results in this study, which are frequently employed for visualizing AD outcomes in the existing literature on HAD. These datasets include six subdatasets from ABU (airport-4, beach-1, beach-3, beach-4, urban-2, and urban-4) and the San Diego dataset. Figs. 7–13 illustrate the graphical detection

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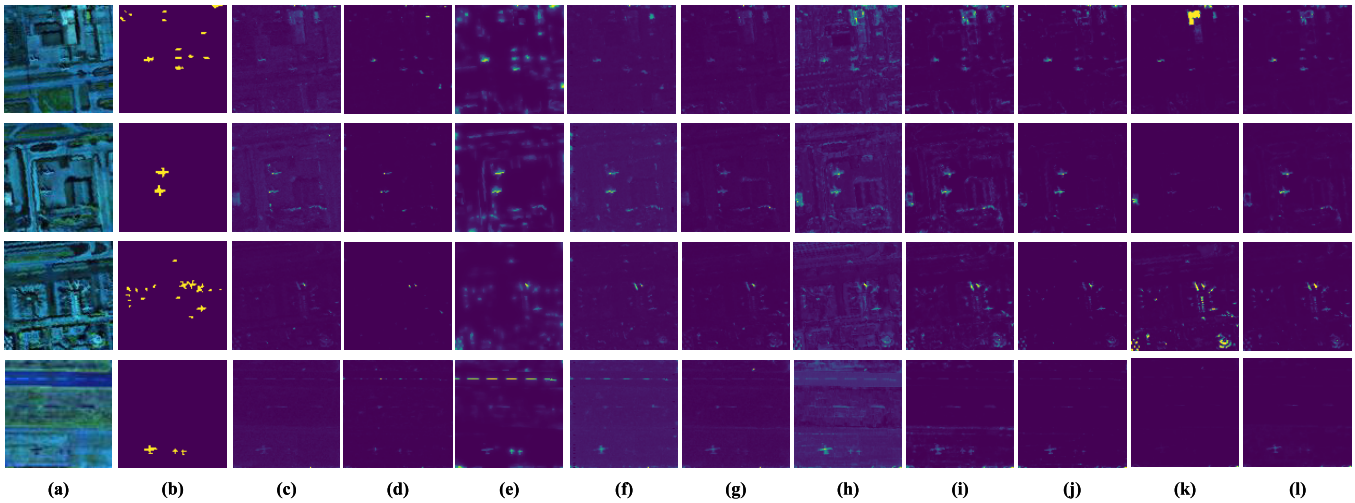


Fig. 3. Detection maps of the compared methods of the airport scene. (a) False color map. (b) Ground truth. (c) RX. (d) LRX. (e) AED. (f) CRD. (g) FrFE. (h) HADGAN. (i) WGAN. (j) BASGAN. (k) RGAE. (l) Proposed method.

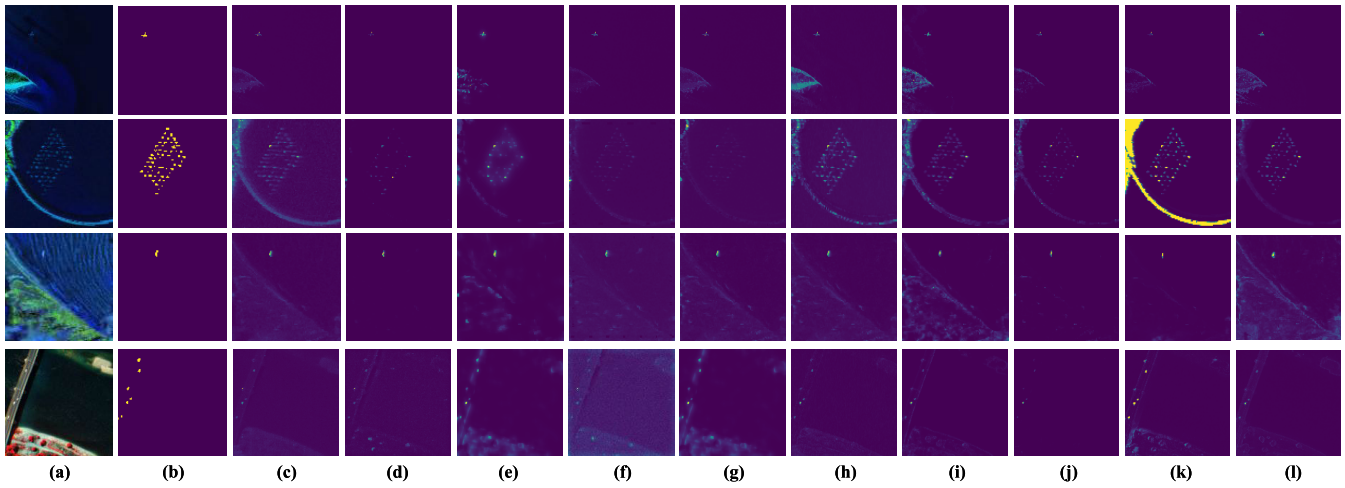


Fig. 4. Detection maps of the compared methods of the beach scene. (a) False color map. (b) Ground truth. (c) RX. (d) LRX. (e) AED. (f) CRD. (g) FrFE. (h) HADGAN. (i) WGAN. (j) BASGAN. (k) RGAE. (l) Proposed method.

results of various algorithms on each dataset, which differ from the existing graphical detection results. To present the detective performance of various algorithms in a more intuitive manner, this article employs a stereo detection result presentation. Specifically, the ground truth and detection results for each dataset are presented as 3-D images, with the lower layer representing the signal strength of each pixel along the Z-axis. The upper layer is a 2-D image that maps the signal strength of the pixels in two dimensions. The pixel color is red when the signal intensity is high, and it is blue when the intensity is low. Figs. 14 and 15 present the 3-D and 2-D ROC curves of ten methods on seven datasets.

The performance of the proposed method on all datasets demonstrates its excellent detection performance, which is superior overall to the other nine methods. The results presented in Figs. 3–6 demonstrate that the proposed method exhibits superior performance in terms of its ability to suppress background pixels and accurately detect anomalous regions. Moreover, the proposed method is able to effectively distinguish the spectral characteristics of anomalous and background image elements, which is a critical requirement for successful AD in remote sensing applications. Table III shows the superior performance of the proposed algorithm in terms of average detection rate compared to other state-of-the-art algorithms. Specifically, the proposed algorithm achieves an average detection rate of 99.32%, which outperforms other methods by margins ranging from 1.44% to 6.78%. In terms of the ability to suppress background interference, Table IV shows that the proposed method ranks third in terms of average false alarm rate but still exhibits better performance than most other algorithms. Compared to the top two algorithms, BASGAN and LRX, the proposed method shows only a minor difference in terms of average false alarm rate. However, the proposed method demonstrates significantly better detection performance, with improvements of 6.78% and 3.20%, respectively. Overall, the proposed method demonstrates outstanding performance in both quantitative evaluation and visual analysis, underscoring its potential for advancing the field of AD.

Specifically, for the airport-4 dataset, the detection results are shown in Fig. 7. The dataset poses a significant challenge

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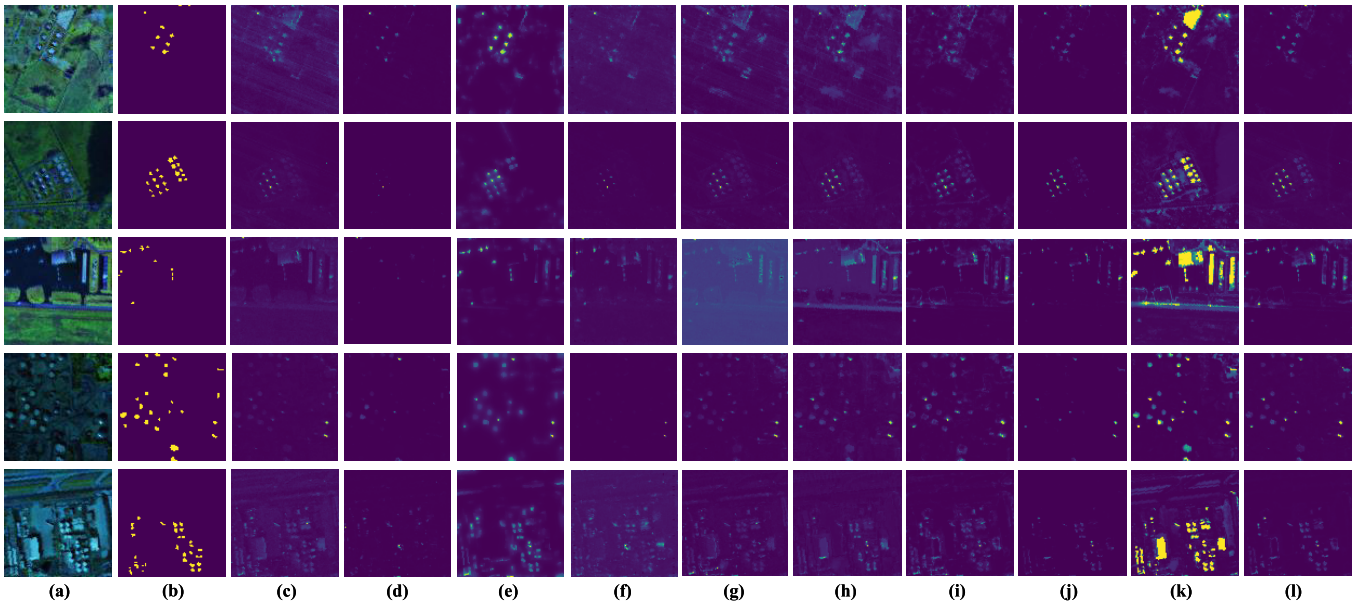


Fig. 5. Detection maps of the compared methods of the urban scene. (a) False color map. (b) Ground truth. (c) RX. (d) LRX. (e) AED. (f) CRD. (g) FrFE. (h) HADGAN. (i) WGAN. (j) BASGAN. (k) RGAE. (l) Proposed method.

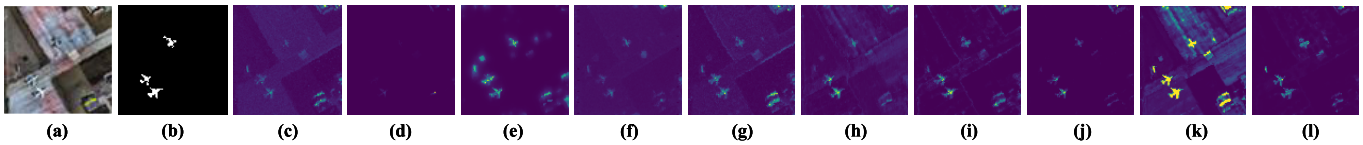


Fig. 6. Detection maps of the compared methods of the San Diego scene. (a) False color map. (b) Ground truth. (c) RX. (d) LRX. (e) AED. (f) CRD. (g) FrFE. (h) HADGAN. (i) WGAN. (j) BASGAN. (k) RGAE. (l) Proposed method.

TABLE III
AUC SCORES OF TPR AND FPR

Datasets	Methods									
	RX	LRX	AED	CRD	FrFE	HADGAN	WGAN	BASGAN	RGAE	Proposed
airport-1	0.8221	0.9549	**0.9922**	0.9229	0.8898	0.9569	0.9517	*0.9624*	0.8894	$\underline{0.9875 \pm 0.0004}$
airport-2	0.8404	0.9217	0.9808	0.9757	0.9515	*0.9849*	0.9830	$\underline{0.9900}$	0.9148	$\mathbf{0.9908 \pm 0.0003}$
airport-3	0.9288	0.9403	$\underline{0.9754}$	0.9442	0.9218	0.9537	0.9514	0.9500	*0.9538*	$\mathbf{0.9852 \pm 0.0017}$
airport-4	0.9536	0.9498	0.9315	0.9650	0.9749	0.9825	$\underline{0.9909}$	*0.9854*	0.9427	$\mathbf{0.9941 \pm 0.0025}$
beach-1	0.9659	0.9736	$\underline{0.9974}$	**0.9977**	0.9666	0.9799	0.9866	0.9705	0.9654	$\mathit{0.9956 \pm 0.0003}$
beach-2	0.9106	0.9353	*0.9551*	0.9224	0.9320	$\underline{0.9857}$	0.9312	0.9421	0.9043	$\mathbf{0.9913 \pm 0.0004}$
beach-3	$\underline{0.9998}$	**1.0000**	*0.9997*	**1.0000**	0.9997	0.9902	0.9981	0.9980	0.9932	$\mathbf{1.0000 \pm 0}$
beach-4	0.9520	0.9313	*0.9792*	0.9259	0.8432	0.9599	$\underline{0.9796}$	0.8849	0.9687	$\mathbf{0.9905 \pm 0}$
urban-1	0.9907	0.9713	$\underline{0.9982}$	0.9880	0.9892	0.9882	0.9906	*0.9975*	0.9843	$\mathbf{0.9989 \pm 0.0001}$
urban-2	*0.9944*	0.6795	0.7953	0.9183	0.9953	0.9967	0.9886	0.9763	$\underline{0.9983}$	$\mathbf{0.9985 \pm 0}$
urban-3	0.9513	0.9272	**0.9976**	0.9239	0.9710	0.9617	0.9745	*0.9850*	0.9777	$\underline{0.9913 \pm 0.0011}$
urban-4	*0.9886*	0.9608	0.8358	0.9527	0.9749	0.9879	0.9699	0.8666	$\underline{0.9934}$	$\mathbf{0.9939 \pm 0.0004}$
urban-5	0.9692	0.8846	*0.9844*	0.9407	0.9617	$\underline{0.9923}$	0.9687	0.9678	0.9636	$\mathbf{0.9931 \pm 0.0007}$
San Diego	0.9403	0.9250	*0.9846*	0.9633	0.9786	0.9822	0.9838	0.9809	$\underline{0.9915}$	$\mathbf{0.9946 \pm 0.0001}$
Average	0.9434	0.9254	0.9577	0.9529	0.9536	$\underline{0.9788}$	*0.9749*	0.9612	0.9601	**0.9932**

to the AD's ability to distinguish between abnormality and background characteristics due to the extensive overlap between the abnormal target spectral curve and some background pixels. The RX, LRX, FrFE, and BASGAN methods' results exhibit varying degrees of abnormal leaks, have a lower response intensity, and fail to preserve the target shape. The AED, CRD, HADGAN, and WGAN methods provide more significant detection results, but background suppression is

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TABLE IV
AUC SCORES OF FPR AND τ

Datasets	Methods									
	RX	LRX	AED	CRD	FrFE	HADGAN	WGAN	BASGAN	RGAE	Proposed
airport-1	0.0442	**0.0064**	0.0347	0.0341	0.0209	0.0666	0.0203	0.0095	0.0175	*0.0101*
airport-2	0.0533	**0.0045**	0.0330	0.0596	0.0187	0.0576	0.0223	*0.0064*	0.0054	0.0106
airport-3	0.0163	**0.0024**	0.0243	0.0268	0.0135	0.0554	0.0192	0.0044	0.0229	*0.0096*
airport-4	0.0266	0.0105	0.0240	0.0682	0.0247	0.0884	0.0126	0.0054	**0.0051**	*0.0092*
beach-1	0.0086	**0.0020**	0.0056	0.0068	0.0088	0.0206	0.0107	0.0032	*0.0044*	0.0093
beach-2	0.0505	**0.0026**	0.0139	0.0229	0.0239	0.0440	0.0187	*0.0080*	0.0905	0.0072
beach-3	0.0278	**0.0024**	0.0135	0.0380	0.0258	0.0508	0.0208	0.0034	*0.0048*	0.0396
beach-4	0.0252	0.0220	0.0217	0.0929	0.3303	0.0171	0.0069	**0.0021**	0.0142	*0.0117*
urban-1	0.0573	0.0065	0.0247	0.0523	0.0444	0.0507	0.0195	**0.0046**	0.2297	*0.0076*
urban-2	0.0155	**0.0022**	0.0160	*0.0092*	0.0146	0.0186	0.0118	0.0025	0.0383	0.0153
urban-3	0.0372	**0.0027**	*0.0155*	0.0306	0.1930	0.0649	0.0187	0.0063	0.5126	0.0201
urban-4	0.0133	*0.0037*	0.0220	0.0026	0.0132	0.0223	0.0132	**0.0022**	0.0151	0.0124
urban-5	0.0455	0.0076	0.0323	0.0631	0.0276	0.0323	0.0196	**0.0038**	0.2258	*0.0083*
San Diego	0.0606	**0.0027**	0.0210	0.0459	0.0507	0.0466	0.0208	0.0046	0.1001	*0.0137*
Average	0.0344	0.0056	0.0216	0.0395	0.0579	0.0454	0.0168	**0.0047**	0.0919	*0.0132*

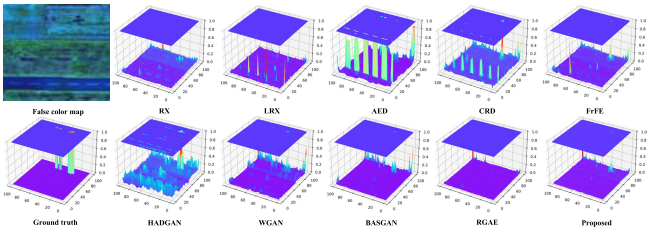


Fig. 7. False color map, ground truth, and detection maps of the compared methods for airport-4.

poor, and the detection result maps contain more background information. The RGAE algorithm eliminates almost all visual anomalies. Despite the airport-4 dataset having few anomalies, the proposed method can accurately identify all the anomalies, better preserve the target shape, and suppress the background. The ROC curve provides a more intuitive comparison of the ten algorithm detection effects. Fig. 14 demonstrates that, compared to the other nine AD algorithms, the proposed method has a higher detection accuracy, even with a low false positive rate, and a higher detection precision. Tables III and IV present the detection rate and the false detection rate of each method on the dataset. The proposed approach achieves an average detection accuracy of 99.41%. Remarkably, this performance metric surpasses all the methods that underwent scrutiny. In addition, the proposed method exhibits a lower false positive rate in contrast to most existing methods.

For the beach-1 dataset, the detection results are shown in Fig. 8. The dataset contains a negligible amount of anomalous information, and the detection algorithm can readily ignore its spectral properties. The LRX algorithm can maintain targets in the ocean with minimal interference from the background, but there is an unusual leakage. The remaining nine algorithms have varying degrees of inaccurate reporting above the right-hand side of the scene, with louder background noise on WGAN. The RX, CRD, FrFE, HADGAN, BASGAN, and RGAE methods suppress sound effects but cannot preserve target edge details. In contrast, the proposed method effectively eliminates interference from the background and accurately identifies anomalies. Table III shows that the proposed method has an average detection accuracy of 99.56%, comparable to the CRD with the highest score. The proposed method has excellent performance in the complete detection of abnormal targets, as evidenced by its comprehensive detection results map and performance evaluation outcomes.

For the beach-3 dataset, the detection results are shown in Fig. 9. This dataset exhibits very small anomalous pixels, and its background spectral characteristics are more diverse, causing strong interference with the AD. Therefore, although the RX, AED, CRD, FrFE, HADGAN, and BASGAN methods can detect anomalies, they are significantly disturbed by the background and produce more erroneous detection results, where the background on the right-hand side of the image is erroneously identified as anomalies. While BASGAN and RGAE are more effective at suppressing the background, they are unable to detect all anomalous targets. The LRX algorithm can suppress background interference more effectively and preserve the targets in the sea relatively intact, but the detection results have weaker response strength. In contrast, the proposed method yields more significant detection results although the background suppression effect is ineffective, and erroneous detections occur. As shown in Table III, the detection performance of each of the ten algorithms is outstanding, and the difference between their values is not statistically significant. The 3-D kurtosis plot shows that the proposed approach has a greater detection intensity for anomalous elements and superior separation from background elements than other detection algorithms. The combined results and

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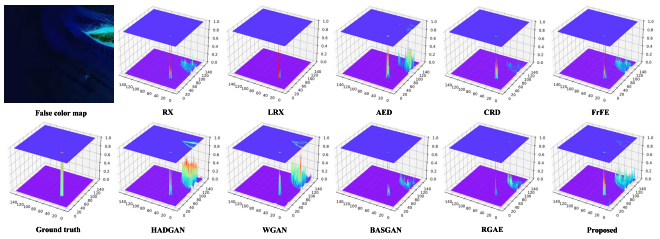


Fig. 8. False color map, ground truth, and detection maps of the compared methods for beach-1.

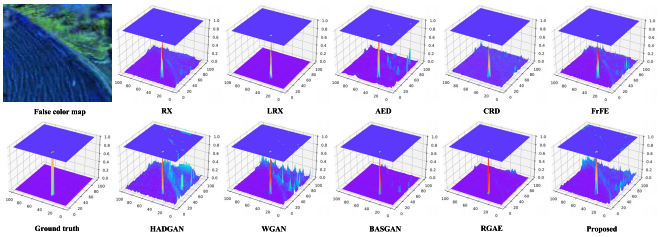


Fig. 9. False color map, ground truth, and detection maps of the compared methods for beach-3.

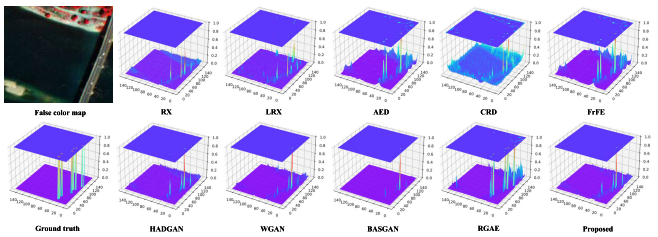


Fig. 10. False color map, ground truth, and detection maps of the compared methods for beach-4.

performance evaluation demonstrate that the proposed method has outstanding performance for detecting anomalies in this dataset.

For the beach-4 dataset, the detection results are shown in Fig. 10. The beach-4 dataset exhibits a sparse distribution of anomalous pixels with extremely tiny pixel sizes. As the water surface dominates the majority of the image, the spectral characteristics of the surrounding bridges and buildings become an unusual component of the dataset, generating more interference for AD. The LRX, AED, CRD, FrFE, and RGAE algorithms produce more false positives and are significantly disturbed by the background, with CRD being the most affected. Meanwhile, the RX, HADGAN, and WGAN algorithms can detect anomalies, but the detection results have a weak response strength and poor subjective perception of the resulting images. The BASGAN algorithm has excellent background suppression and fewer false positive detections, but it ignores some outliers. The 3-D kurtosis plot indicates that the proposed method outperforms other algorithms in terms of background suppression while maintaining a high level of anomalous region detection sensitivity. The $AUC_{(TPR,FPR)}$ values for the proposed method and the nine compared algorithms are presented in Table III. The detection accuracy of the proposed algorithm is the highest among all algorithms on the beach-4 dataset, with $AUC_{(TPR,FPR)}$ of 99.05%. The proposed method's detection accuracy is 1.09% higher than the second-place WGAN and 14.73% higher than the lowest FrFE algorithm. Table IV shows that BASGAN, WGAN, and the proposed method perform the best at suppressing background interference. Although the proposed method did not achieve the highest false detection performance, its intuitive detection performance is superior to that of the majority of advanced algorithms. By combining detection result graphs and performance evaluation results, the proposed method enhances the performance of AD on this dataset.

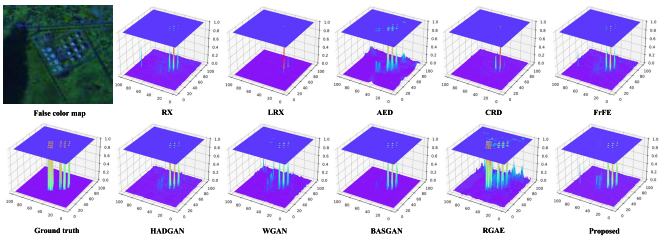


Fig. 11. False color map, ground truth, and detection maps of the compared methods for urban-2.

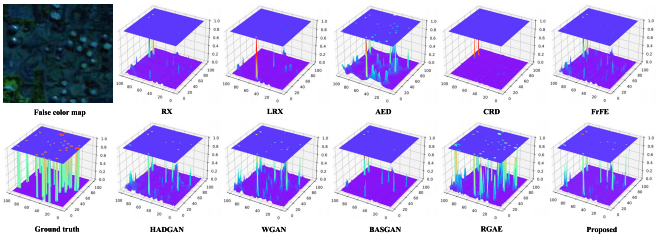


Fig. 12. False color map, ground truth, and detection maps of the compared methods for urban-4.

For the urban-2 dataset, the detection results are shown in Fig. 11. Regarding subjective effects, the RX, AED, FrFE, HADGAN, and WGAN algorithms produce comparable detection outcomes, which are significantly influenced by the background. The LRX, CRD, and BASGAN algorithms have exceptional background suppression and fewer false positives, but there are anomalous absences. The RGAE algorithm yields more significant detection results, but it has weak background suppression and a greater number of false positives. In contrast, the presented method can effectively detect anomalies and describe the geometry of the anomalous target in further detail. As shown in Table III, the proposed approach achieves the best $AUC_{(TPR,FPR)}$ while obtaining better background suppression effect. Compared with other algorithms, the average detection rate of the method has increased by 7.15%. In terms of false detection performance, although the S2G2HAD's false detection performance is not the lowest, it is generally satisfactory. In conclusion, the proposed method performs excellently on the urban-2 dataset.

For the urban-4 dataset, the detection results are shown in Fig. 12. The RX, FrFE, HADGAN, and WGAN algorithms erroneously identify a portion of the background as anomalous, whereas LRX, CRD, and BASGAN exhibit partial anomaly misses such that the target's shape is not entirely preserved. The AED algorithm can effectively highlight the anomalous target but cannot precisely define the shape of the anomaly. The RGAE algorithm can detect anomalies adequately but cannot effectively suppress background noise. In comparison

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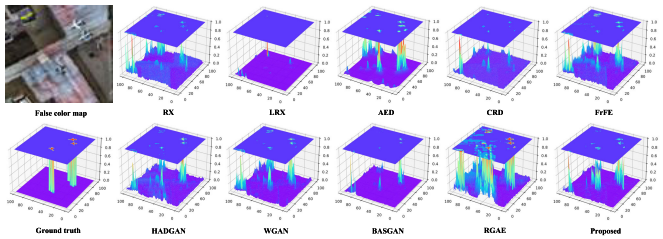


Fig. 13. False color map, ground truth, and detection maps of the compared methods for San Diego.

to other detection algorithms, the proposed approach yields the most accurate detection results. The numerical analysis in Tables III and IV indicates that the average $AUC_{(TPR,FPR)}$ of the proposed method is 99.39%, which is 0.05% higher than the RGAE algorithm with the second-highest $AUC_{(TPR,FPR)}$. The detection rate of the proposed method is on average 4.61% higher than that of other algorithms. Although the LRX, CRD, and BASGAN algorithms obtain a low $AUC_{(FPR,\tau)}$ for the false detection performance evaluation, they suffer from significant leakage issues at the expense of detection accuracy. Overall, it appears that the proposed method outperforms other advanced algorithms in terms of detection performance.

For the San Diego dataset, the detection results are shown in Fig. 13. Due to the similar geomaterials of some background areas (e.g., hangars) and anomalous targets (aircraft), accurate target detection is challenging. RX, AED, CRD, FrFE, HADGAN, WGAN, and RGAE algorithms fail to completely separate the anomalies from the background and produce a high number of false positives. LRX and BASGAN algorithms suppress the background more effectively but at the cost of missed detection. The proposed method not only precisely identifies the locations of the three aircraft (anomaly targets) but also defines their entire shapes. In comparison to other anomaly algorithms, the proposed approach yields the most visually similar result to the reference map. The ROC curve indicates that the proposed method's ROC curve is generally higher than the ROC curves of other algorithms, indicating that the proposed method's detection effect is superior to those of other algorithms. When the detection probability of each algorithm reaches 100%, the false alarm rate of the proposed method is the lowest among all algorithms, which is demonstrably superior to other algorithms. With an average $AUC_{(TPR,FPR)}$ of 99.46%, the proposed method achieves the highest detection accuracy on the San Diego dataset among all other algorithms. Compared to other algorithms, the proposed method improves the average detection accuracy by 2.46%. Table IV shows that the proposed method is not the most effective at suppressing background interference, but its false detection performance rates third. Compared to the LRX and BASGAN algorithms, which have exceptional background suppression in this dataset, the proposed method's detection accuracy is improved by 7.00% and 1.37%, respectively. Based on the preceding analysis, it can be confirmed that the proposed method is a highly competitive method for AD.

D. Ablation Study

Ablation study is an important method for evaluating the rationality and effectiveness of algorithms. To assess the contribution of the spatial–spectral feature dynamic composition module and the graph-guided Siamese selective reconstruction module to the AD performance, two ablation studies are performed in the ABU and San Diego datasets, respectively.

1) Effectiveness of the Spatial–Spectral Feature Dynamic Composition Module: In this article, we propose the spatial–spectral feature dynamic composition module, which incorporates the three-way clustering theory to acquire pseudolabels of image pixels in the absence of prior information. The proposed approach further divides the original data into "background," "anomaly," and "uncertain" classes, in addition to the conventional "background" and "anomaly" classes. This strategy enables more accurate feature embedding and graph construction in scenarios where prior information is unavailable, outperforming conventional approaches that only divide the data into two classes. To assess the impact of spatial–spectral feature dynamic composition module on the performance of the method, relevant ablation experiments are designed and conducted, aimed at providing insights into the effectiveness of the proposed approach and its potential for real-world applications. This study conducts two groups of experiments to investigate the impact of the spatial–spectral feature dynamic composition module. In Experiment 1, the pseudolabels of image pixels are simply marked as "background" and "anomaly" before the subsequent detection work. In Experiment 2, the proposed spatial–spectral feature dynamic composition module, which integrates the three-way clustering theory, is introduced to divide the pixels into "background," "anomaly," and "uncertain" classes, with the aim of improving the accuracy of feature embedding and graph construction. The detection AUCs of Experiment 1 and Experiment 2 are compared to evaluate the effectiveness of the spatial–spectral feature dynamic composition module in the proposed method. The detection results of Experiment 1 and Experiment 2 are presented in Fig. 16(a) and (b), which indicates that the proposed module can effectively enhance the detection performance on all datasets. Specifically, the performance of the method is improved by 0.61%, 0.39%, 0.61%, 0.34%, 0.24%, 0.37%, 0.11%, 0.57%, 0.22%, 0.19%, 0.32%, 0.40%, 0.38%, and 0.25% in each dataset, with an average increase of 0.36%. These findings demonstrate that the proposed module enables more accurate feature embedding and graph construction in scenarios where prior information is unavailable, thereby effectively enhancing the detection performance of the proposed method.

2) Effectiveness of the Graph-Guided Siamese Selective Reconstruction Module: To evaluate the efficacy of the graph-guided Siamese selective reconstruction module in the proposed method, a set of experiments are conducted. Experiment 1 replaces the S2BAE with a single GATE for background reconstruction and AD tasks, while Experiment 2 uses the original Siamese network for the same tasks. The detection AUC and runtime of Experiment 1 are compared with that of Experiment 2 to verify the role of the S2BAE in the proposed method. Furthermore, the detection AUC and runtime of Experiment 2 are compared with that of the proposed method in Tables III and VI to investigate the

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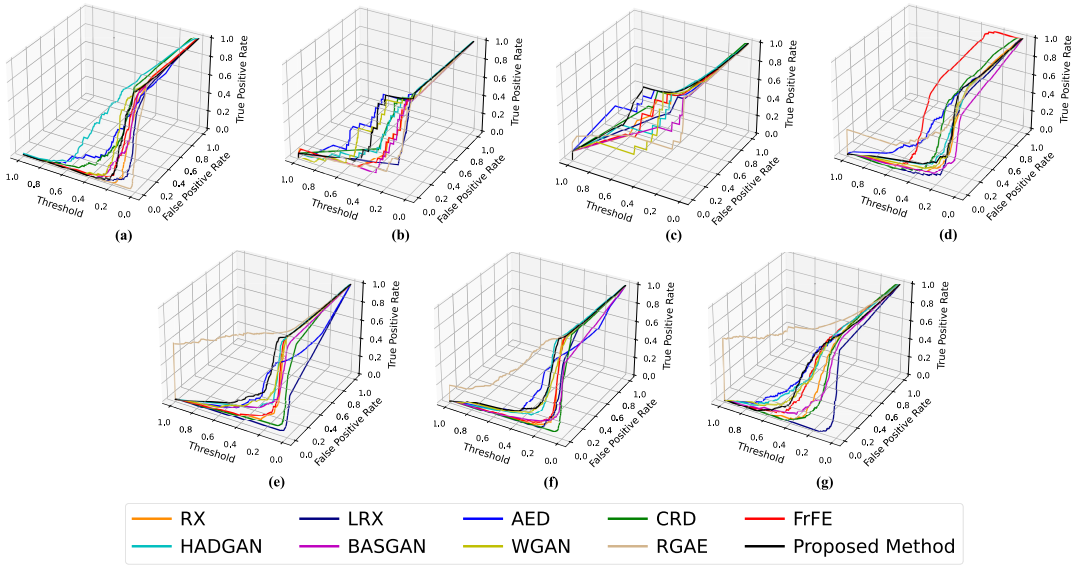


Fig. 14. 3-D ROC curves for the ten methods on (a) airport-4, (b) beach-1, (c) beach-3, (d) beach-4, (e) urban-2, (f) urban-4, and (g) San Diego.

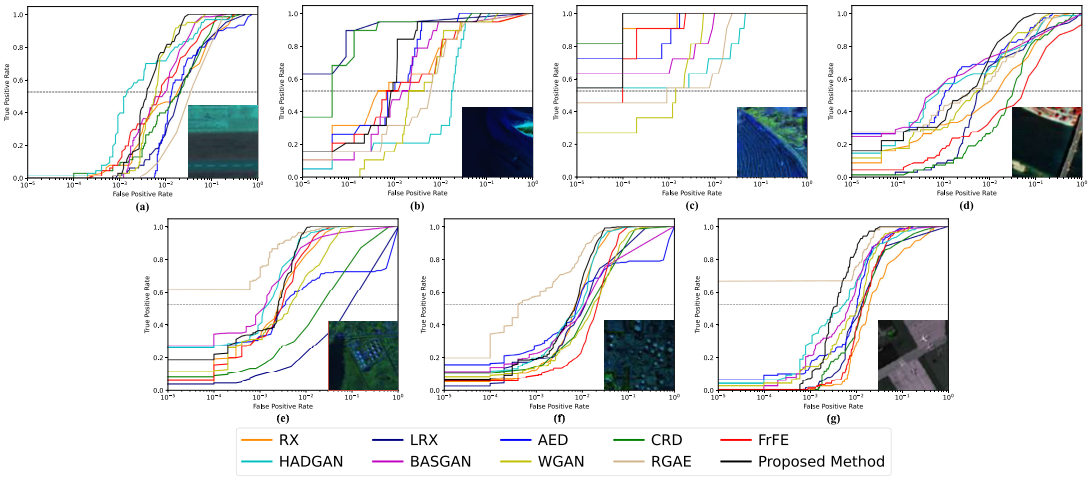


Fig. 15. 2-D ROC curves of (TPR, FPR) for the ten methods on (a) airport-4, (b) beach-1, (c) beach-3, (d) beach-4, (e) urban-2, (f) urban-4, and (g) San Diego.

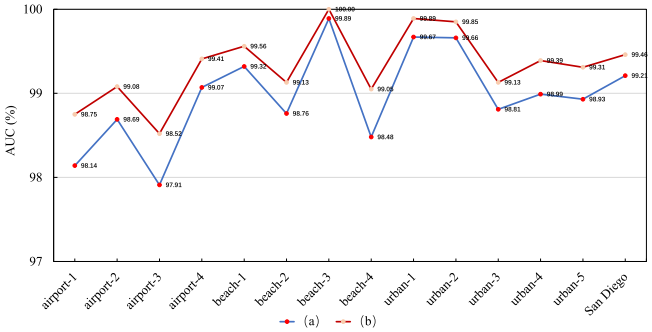


Fig. 16. Ablation study of the spatial–spectral feature dynamic composition module using the metric of AUC on (a) anomaly detection under binary pseudo-labels and (b) anomaly detection under ternary pseudo-labels.

roles of the CEDm in the proposed method. The detection results of Experiment 1 and Experiment 2 are presented in Table V(a) and (b), respectively. For comparison purposes, Table V(c) displays the detection AUC performance and runtime of the proposed method, which are consistent with the values reported in Tables III and VI.

The experimental results are shown in Table V. As can be seen from the table, when using only a single GATE, the AUC of the method has reached 0.9798, 0.9834, 0.9778, 0.9889, 0.9883, 0.9885, 0.9960, 0.9846, 0.9923, 0.9943, 0.9848, 0.9880, 0.9839, and 0.9899 on the datasets, respectively. After using two GATEs with the same parameters to construct the S2BAE, the performance of the method has been improved by 0.38%, 0.32%, 0.23%, 0.44%, 0.58%, 0.15%, 0.22%, 0.54%, 0.39%, 0.39%, 0.27%, 0.43%, 0.49%, and 0.51% in each group of data. After introducing CEDm, the detection performance is further improved by 0.39%, 0.42%, 0.51%, 0.08%, 0.15%, 0.13%, 0.18%, 0.05%, 0.27%, 0.03%, 0.38%, 0.16%, 0.43%, and 0.33%. The results indicate that each module used in the entire reconstruction process contributes to the improvement of AD performance. In terms of computational efficiency, the original reconstruction mechanism based on the Siamese network incurs a slight increase in

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TABLE V
ABLATION STUDY OF THE GRAPH-GUIDED SIAMESE SELECTIVE RECONSTRUCTION MODULE USING THE METRIC OF AUC AND TIME (IN SECONDS)

Datasets	Configuration		
	(a)	(b)	(c)
airport-1	0.9798/0.1709	0.9836/0.2071	0.9875/0.2011
airport-2	0.9834/0.1499	0.9866/0.1816	0.9908/0.1764
airport-3	0.9778/0.1489	0.9801/0.1804	0.9852/0.1752
airport-4	0.9889/0.1796	0.9933/0.2176	0.9941/0.2113
beach-1	0.9883/0.3175	0.9941/0.3847	0.9956/0.3735
beach-2	0.9885/0.1533	0.9900/0.1857	0.9913/0.1803
beach-3	0.9960/0.1540	0.9982/0.1866	1.0000/0.1812
beach-4	0.9846/0.1680	0.9900/0.2036	0.9905/0.1977
urban-1	0.9923/0.1466	0.9962/0.1776	0.9989/0.1725
urban-2	0.9943/0.1510	0.9982/0.1830	0.9985/0.1777
urban-3	0.9848/0.1449	0.9875/0.1756	0.9913/0.1705
urban-4	0.9880/0.1751	0.9923/0.2121	0.9939/0.2060
urban-5	0.9839/0.1576	0.9888/0.1909	0.9931/0.1854
San Diego	0.9862/0.1442	0.9913/0.1746	0.9946/0.1696
Average	0.9869/0.1687	0.9907/0.2043	0.9932/0.1985

runtime compared to using a single GATE, with an average increase of 0.0344 s. However, with the incorporation of CEDm, the average runtime decreases by 0.0071 s compared to the original reconstruction mechanism based on the Siamese network. By conducting a comprehensive evaluation of both detection performance and runtime, it can be concluded that the graph-guided Siamese selective reconstruction module effectively enhances the spatial perception capability of the model, reduces the search space and execution time for each pixel, and improves the purity of reconstructed backgrounds. Consequently, this leads to an improvement in the accuracy of AD.

E. Parameter Analyses

The analysis of model parameters is essential for assessing model reliability, identifying crucial parameters, and optimizing model performance. Hence, this study conducted parameter analysis experiments on pivotal parameters, namely, superpixel number, latent vector dimension, random walk sampling number, margin, and feature identification threshold, using the San Diego dataset to evaluate the rationality and effectiveness of parameter selection.

1) Evaluations Under Different Superpixel Numbers: The proposed method employs superpixel segmentation to construct graph data, and the number of superpixels used has a significant impact on the performance. To this end, an experiment is designed to investigate the effect of varying superpixel numbers on the performance and determine the optimal number of superpixels for the proposed method. Specifically, we evaluate the AD performance under ten different superpixel numbers (20, 40, 60, 80, 100, 200, 300, 400, 500, and 600) and analyze the results, as shown in Fig. 17(a). The results indicate that the detection performance initially improves with an increase in the number of superpixels and then reaches a peak before decreasing. The optimal detection performance is observed when using 100 superpixels, with an AUC score of 99.46%. The experimental results also suggest that a higher number of superpixels generally result in better network performance. Nevertheless, a large number of superpixels can increase computational complexity, training and inference time, and resource consumption. Furthermore, the network may focus excessively on data details and overlook overall features, which can lead to overfitting if the number of superpixels is too large. Based on these considerations and to ensure experiment fairness, we set the number of superpixels to 100.

2) Evaluations Under Different Latent Vector Dimension: This study investigates the impact of latent vector dimensions on computation efficiency by conducting a parameter analysis experiment involving varying latent vector dimensions. The objective of the experiment is to assess the influence of latent vector dimensions on AD performance. The results of the experiments are depicted in Fig. 17(b), where the horizontal axis represents the latent vector dimension, while the vertical axis represents the AUC score, which assesses the AD performance. The results indicate that computation performance initially improves as the dimension of the latent vector increases but then decreases. The optimal detection performance is achieved at a latent vector dimension of 64. Further analysis of the experimental results reveals that latent vectors map input data into a low-dimensional space, extracting key features from the data and converting high-dimensional data into a low-dimensional vector representation. However, if the latent vector dimensions are too large, the models may overfit the noise and detail information in the training data while neglecting the more important features. To ensure experiment fairness, we set the latent vector dimension to 64.

3) Evaluations Under Different Random Walk Sampling Numbers: A parameter analysis experiment is conducted to investigate the impact of random walking sampling on method efficacy. The experiment aims to determine the optimal number of random wandering samples for detecting anomalies. The results of the experiment are presented in Fig. 17(c), where the horizontal axis represents the number of random wandering samples in the range [100, 800] with an interval of 100, while the vertical axis represents the corresponding AUC performance. The figure shows that the AD performance initially improves as the number of random walk samples increases from 100 to 400, while the AUC remains constant as the number of random walk samples continues to increase. Based on the processing time considerations, we select 400 random wandering samples. Further analysis of the experimental results indicates that the number of random wandering samples corresponds to the number of samples used for model training. Large-scale datasets can theoretically provide more comprehensive and detailed information to assist the model in learning more accurate and generalized features. However, the computational resources consumed by the training dataset need to be evaluated. Therefore, considering overall factors

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TABLE VI
AVERAGE COMPUTING TIME (IN SECONDS) OF THE COMPARATIVE METHODS

Datasets	Methods									
	RX	LRX	AED	CRD	FrFE	HADGAN	WGAN	BASGAN	RGAE	Proposed
airport-1	0.2194	54.7945	0.3089	16.8180	16.8495	*0.2796*	0.3019	0.2799	81.3244	**0.2011**
airport-2	0.2049	51.8163	0.3478	16.9014	15.8498	*0.2645*	0.2946	0.2648	64.4894	**0.1764**
airport-3	0.1876	52.0635	0.3697	17.4785	15.2362	*0.2558*	0.3085	0.2862	59.8198	**0.1752**
airport-4	**0.1956**	42.9308	0.2814	16.9139	14.5476	*0.2736*	0.3074	0.2994	52.8160	0.2113
beach-1	*0.3562*	90.4966	0.4421	39.5611	30.4066	**0.2633**	0.2893	0.5233	132.2525	0.3735
beach-2	0.2688	45.1223	0.3944	16.4294	16.9498	0.2712	*0.2568*	0.2549	63.4894	**0.1803**
beach-3	*0.2462*	40.6670	0.3594	16.8586	13.9830	0.2648	0.2387	0.2655	57.8077	**0.1812**
beach-4	0.2071	37.7053	0.4073	34.9590	20.7185	*0.2545*	0.2719	0.2704	104.8476	**0.1977**
urban-1	0.1947	47.5262	0.2713	17.0172	14.6841	0.2841	0.2922	*0.2475*	61.8853	**0.1725**
urban-2	0.2065	53.0829	0.2738	17.1392	16.8846	0.2765	*0.2663*	0.2801	77.6958	**0.1777**
urban-3	0.2211	41.8603	0.2541	16.2786	13.7409	0.2669	0.2938	*0.2315*	57.9484	**0.1705**
urban-4	**0.1733**	142.8728	0.2606	17.2066	15.3297	0.2955	0.2876	*0.2471*	77.0959	0.2060
urban-5	**0.1647**	54.1125	0.2907	17.2730	16.6184	0.2683	0.3313	*0.2493*	78.9196	0.1854
San Diego	0.1708	129.2281	0.2587	17.2016	13.4973	0.2406	0.3656	*0.2337*	79.4328	**0.1696**
Average	0.2155	63.1628	0.3229	19.8597	16.8069	*0.2685*	0.2933	0.2810	74.9875	**0.1985**

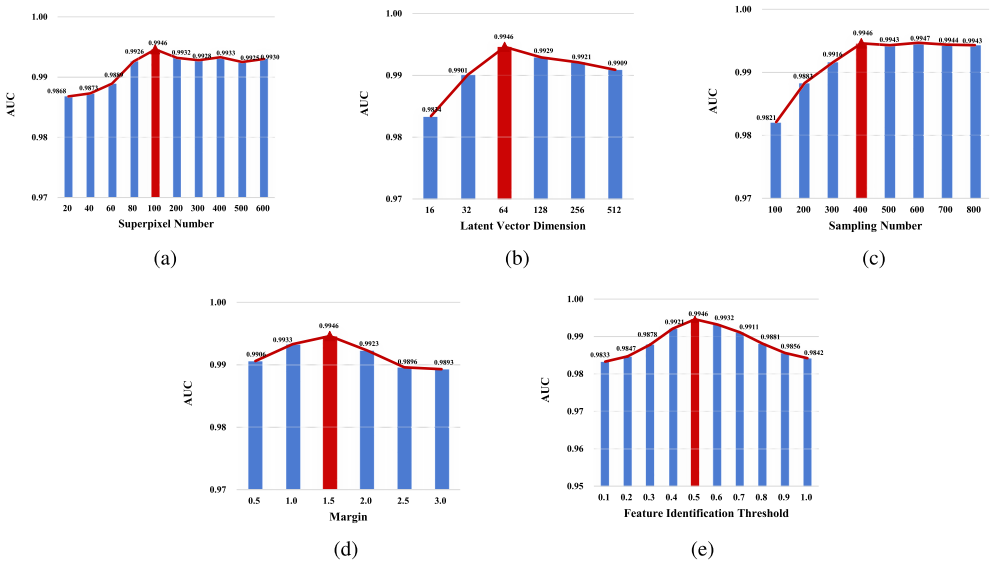


Fig. 17. Analysis of the parameters on the San Diego dataset. (a) Superpixel number. (b) Latent vector dimension. (c) Random walk sampling number. (d) Margin. (e) Feature identification threshold.

and ensuring the fairness of the entire experiment, we set the number of random walk samples to 400 during the experiment.

4) Evaluations Under Different Margins: A validation experiment is conducted to investigate the impact of the margin parameter on method performance. This experiment involves varying margin values to obtain the corresponding AD performance, as shown in Fig. 17(d). The horizontal axis in Fig. 17(d) represents the margin value, which ranges from 0.5 to 3 with increments of 0.5, while the vertical axis represents the corresponding AUC performance. The optimal AUC performance is achieved when the margin is set to 1.5. The margin parameter plays a crucial role in controlling the distance between similar and dissimilar sample pairs, preventing the network from compressing the distance between all sample pairs, which can lead to an inability to distinguish between similar and dissimilar pairs. In practical applications, a margin value of 1.5 is selected as it balances class intracoherence and interclass differences, providing a reasonable balance point for model optimization and classification performance. To ensure the fairness of the entire experiment, the margin value is set to 1.5 during the experiment.

5) Evaluations Under Different Feature Identification Threshold: A verification experiment systematically explores the influence of the feature identification threshold on algorithmic performance, as visually represented in Fig. 17(e). Within this graphical representation, the horizontal axis delineates

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the numerical spectrum of the feature identification threshold, spanning from 0 to 1 in increments of 0.1, while the vertical axis delineates the corresponding performance metrics in terms of the AUC for AD. As the threshold increases, algorithm performance initially rises and then declines, reaching optimal performance at 0.5. The threshold aims to balance data purity and ambiguity reduction. High values result in loss of information and reduced performance, while low values lead to the mixing of spectral characteristics, causing more false negatives. Setting the threshold to 0.5 achieves the best equilibrium between data purity and filtering. Thus, for consistency, the threshold is maintained at 0.5 throughout the experiments.

F. Model Complexity Analysis

1) Time Complexity Evaluation: Time complexity is a crucial indicator of algorithm efficiency. Therefore, this section analyzes the time complexity of each method by measuring the execution time of various algorithms on the ABU and San Diego datasets to validate their efficacy. The experimental results are presented in Table VI and Fig. 18 where Table VI displays the execution durations of the ten methods on all hyperspectral datasets in seconds, and Fig. 18 provides a visual representation of the runtime performance of different algorithms across all datasets. Based on the results presented in Table VI, it can be observed that the proposed method achieves the most efficient detection performance on all datasets, except for airport-4, beach-1, urban-4, and urban-5. On the airport-4, urban-4, and urban-5 datasets, the proposed method performs slightly worse than the RX algorithm, with differences in runtime of 0.0157, 0.0327, and 0.0207 s, respectively. On the beach-1 dataset, the proposed method achieves comparable runtime efficiency with HADGAN, WGAN, and the RX algorithm. Moreover, when combined with the data presented in Table III, the proposed method outperforms other methods with shorter runtime (RX, HADGAN, and WGAN) in terms of detection accuracy. In addition, as demonstrated in the last column of Table VI, the proposed method ranks first in terms of average runtime, with an average reduction of 19.38 s compared to other methods. Thus, it can be concluded that the proposed method achieves high detection accuracy while satisfying low time costs, meeting the timeliness requirement of HAD.

2) Spatial Complexity Evaluation: Spatial complexity is a significant indicator of model lightness. Accordingly, this section assesses the spatial complexity of each method by examining the relationship between the number of network parameters, running time, and detection performance of various methods on the San Diego dataset. The experimental results are presented in Fig. 19, where the horizontal axis represents the detection time in seconds, the vertical axis represents the AUC score that measures the detection performance, and the circle's diameter represents the number of network parameters, with a larger diameter indicating a greater number of network parameters. As the RX, LRX, AED, CRD, FrFE, and RGAE algorithms do not construct a neural network structure, they are not included in this metric's comparison,

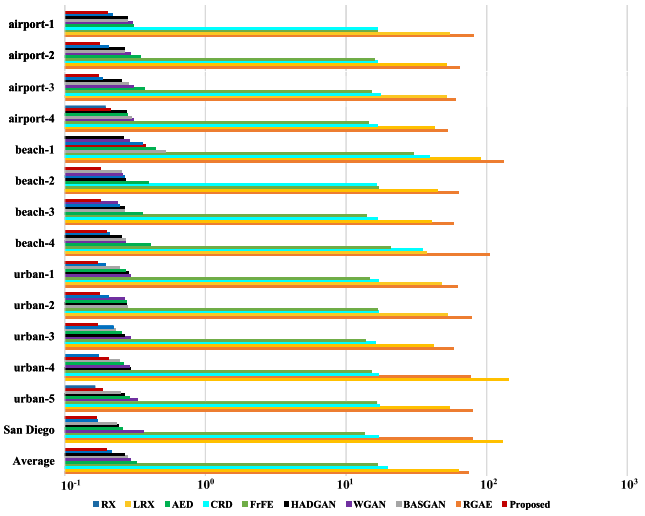


Fig. 18. Time complexity visualization for the ten methods.

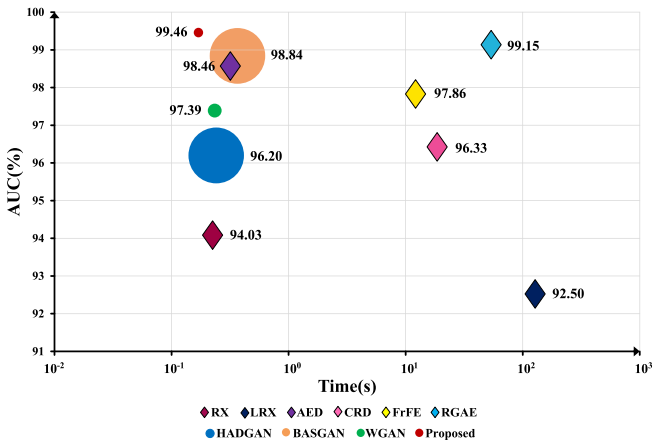


Fig. 19. Space complexity visualization of the compared methods for San Diego.

and only the detection time and detection performance, represented by a diamond in the figure, are evaluated. Fig. 19 provides a visual comparison of the detection precision and inference performance of the ten methods. The experimental results indicate that the proposed method achieves a state-of-the-art balance between accuracy and speed, with a detection accuracy of 99.46% and an inference speed of 0.1696 s. With a parameter count of 0.07M, the method presented in this article achieves model lightening while providing the fastest inference speed and highest detection accuracy. The above analysis confirms that the proposed method meets the requirement for model lightweighting.

V. Conclusion

This article proposes an innovative S2G2HAD method. To begin, the spatial–spectral feature dynamic composition module enhances mutual information among data elements using hyperspectral graph data construction. In addition, a three-way clustering-guided strategy effectively separates category-ambiguous data, reducing its impact on subsequent precision. Then, the CEDm is introduced, incorporating

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similarity discrimination from the Siamese network framework into the hidden layers to enhance the encoder's feature extraction and differentiation capabilities. This aids in explicitly separating hidden features of heterogeneous data in high-dimensional space, ultimately improving the purity of background spectral reconstruction. Finally, the graph-guided Siamese selective reconstruction module is utilized to differentially extract graph data features from structurally enriched graph data. Prioritizing the use of background graph features for pristine background reconstruction, it accentuates interconnections among pixel samples, establishing cross-spectral band associations. This strategic approach efficiently removes redundancy, enhancing data processing efficiency and reducing computational resource consumption. Extensive experiments on seven international public datasets demonstrate that the proposed method outperforms nine state-of-the-art methods in terms of detection performance and efficiency.

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